Design of the Symmetrical OTA

For Generic 180nm Bulk CMOS Process (Version 3)

Christian Enz (christian.enz@epfl.ch)

24.09.2026

This notebook presents the analysis, design and simulation of the symmetrical cascode OTA.

Table of contents

1 Introduction **4**

2 Analysis **5**
2.1 Small-signal analysis . . . 5
2.2 Noise analysis . . . 7
2.2.1 Input-referred thermal noise . . . 8
2.2.2 Input-referred flicker noise . . . 8
2.3 Input-referred offset voltage . . . 9

3 Design **11**
3.1 Specifications . . . 11
3.2 Process . . . 11
3.3 Design procedure . . . 13
3.3.1 Sizing the differential pair M_{1a}-M_{1b} . . . 13
3.3.2 Sizing the pMOS current mirrors M_{2a}-M_{2c} and M_{2b}-M_{2d} . . . 14
3.3.3 Sizing the nMOS current mirror M_{3a} and M_{3b} . . . 16
3.3.4 Sizing the cascode transistors M_4-M_7 . . . 19
3.3.5 Sizing the nMOS current mirror M_{5a}-M_{5b} . . . 22
3.3.6 Final sizing of the differential pair M_{1a}-M_{1b} . . . 22
3.4 Summary . . . 23
3.4.1 Specifications . . . 23
3.4.2 Bias . . . 24
3.4.3 Transistor information . . . 24

4 OTA Characteristics **26**
4.1 Open-loop gain . . . 26
4.2 Input-referred noise . . . 26
4.3 Input-referred offset voltage . . . 28
4.4 Current and power consumption . . . 30

5 Simulation results from ngspice **31**
5.1 Operating point . . . 31
5.2 Large-signal differential transfer characteristic . . . 35
5.3 Open-loop gain . . . 36
5.3.1 Closed-loop circuit . . . 36
5.3.2 Open-loop circuit . . . 36
5.4 Input-referred noise . . . 38
5.5 Input common-mode voltage range . . . 40
5.6 Step-response . . . 41
5.6.1 Small-step . . . 41
5.6.2 Large step . . . 41
5.7 Current and power consumption . . . 43

6 Conclusion **44**

References **45**

© C. Enz under CC BY-NC-SA

1 Introduction

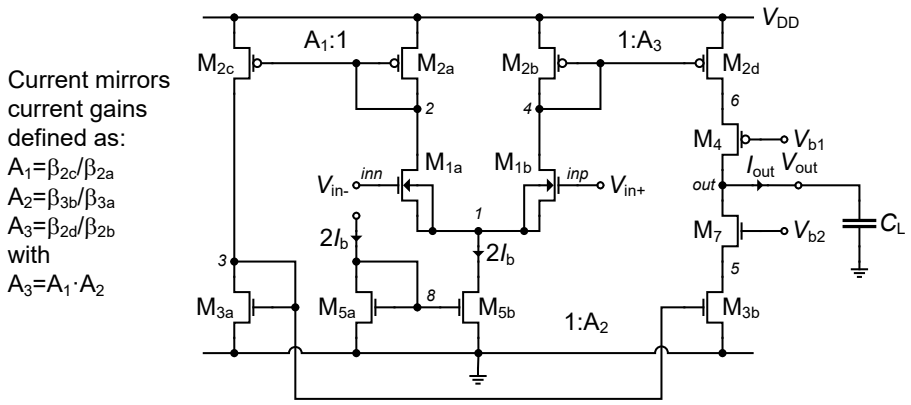


Figure 1.1: Schematic of the symmetrical differential OTA.

> **Note**
>
> Note that all nMOS transistors in Figure 1.1 have an odd number and all the pMOS transistors an even number which explains the numbering process (this is the reason why there is no M_6 transistor!).

This notebook presents the analysis, design and simulation of the symmetrical cascode OTA which schematic is presented in Figure 1.1 [1]. The design phase is using the sEKV model and the inversion coefficient approach [2], [3], [4]. The symmetrical OTA is a single-stage OTA having the dominant pole set by the load capacitance. It therefore doesn't require any compensation capacitance but can still achieve the gain of a two-stage OTA thanks to the cascode stages at the output at much lower current consumption. We will see below that in differential mode the effects of the source transconductances on the common source node voltage are actually canceled. In order to minimize the V_{GS} voltage of M_{1a}-M_{1b}, we have chosen to put M_{1a}-M_{1b} in a separate well at the cost of a larger area.

We will start with a detailed analysis of the OTA which will allow to derive all the design equations that will be used in the design phase. The OTA is then designed for a given set of specifications for a generic 180 nm bulk CMOS technology. The design is then validated by simulations with ngspice [5] using the EKV 2.6 compact model [6] [7] [8] with parameters corresponding to a generic 180 nm bulk CMOS technology [9] [10].

We now start with the small-signal analysis.

© C. Enz under CC BY-NC-SA

2 Analysis

2.1 Small-signal analysis

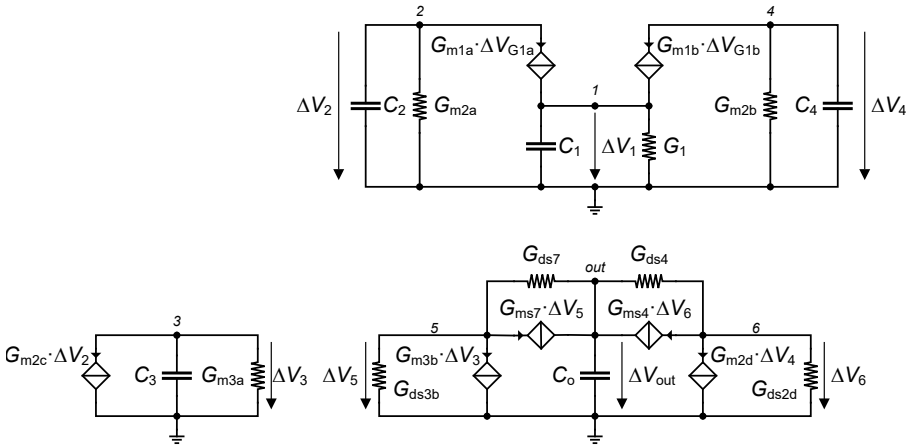


Figure 2.1: Small-signal schematic of the symmetrical cascode OTA.

The small-signal schematic of the symmetrical OTA of Figure 1.1 is shown in Figure 2.1. The design of the symmetrical cascode OTA is similar to the design of the simple OTA, except that there are now more parasitic poles appearing at the various current mirror nodes. The small-signal output voltage in open-loop is given by

$$\Delta V_{out} = \frac{A_{dc}}{1+s\,\tau_0} \cdot \left[\frac{\Delta V_{in+}}{1+s\,\tau_{p4}} - \frac{\Delta V_{in-}}{(1+s\,\tau_{p2})(1+s\,\tau_{p3})} \right. \tag{2.1}$$

$$\left. - \frac{\Delta V_{in+} + \Delta V_{in-}}{2(1+s\,\tau_{p1})} \cdot \left(\frac{1}{1+s\,\tau_{p4}} - \frac{1}{(1+s\,\tau_{p2})(1+s\,\tau_{p3}} \right) \right] \tag{2.2}$$

where

$$A_{dc} = \frac{G_m}{G_o}, \tag{2.3}$$

$$G_m = A_3 \cdot G_{m1}, \tag{2.4}$$

$$G_o = \frac{G_{ds3b}\,G_{ds7}}{G_{ms7}} + \frac{G_{ds2d}\,G_{ds4}}{G_{ms4}}, \tag{2.5}$$

$$\tau_0 = \frac{C_{out}}{G_o}, \tag{2.6}$$

$$C_{out} = C_L + C_o, \tag{2.7}$$

$$\tau_{p1} = \frac{C_1}{2G_{m1}}, \tag{2.8}$$

$$\tau_{p2} = \frac{C_2}{G_{m2}}, \tag{2.9}$$

$$\tau_{p3} = \frac{C_3}{G_{m3}}, \tag{2.10}$$

$$\tau_{p4} = \frac{C_4}{G_{m2}}. \tag{2.11}$$

© C. Enz under CC BY-NC-SA

C_1, C_2 and C_3 are the total parasitic capacitances at the current mirrors nodes 1, 2 and 3.

The unity-gain frequency or gain-bandwidth product is defined as $\omega_u = 1/\tau_u \triangleq A_{dc}/\tau_0 = G_m/C_{out}$. In most practical cases, we usually have $\tau_0 \gg \tau_{p2}, \tau_{p3}, \tau_{p4} > \tau_{p1}$.

For a differential input voltage $\Delta V_{id} = \Delta V_{in+} - \Delta V_{in-}$ with a constant common-mode voltage V_{ic} and hence $\Delta V_{ic} = (\Delta V_{in+} + \Delta V_{in-})/2 = 0$, $\Delta V_{in+} = -\Delta V_{in-} = \Delta V_{id}/2$, the differential mode small-signal open-loop gain reduces to

$$A_{dm}(s) \triangleq \frac{\Delta V_{out}}{\Delta V_{id}} = \frac{1}{2} \cdot \frac{A_{dc}}{1 + s\,\tau_0} \cdot \left[\frac{1}{1 + s\,\tau_{p4}} - \frac{1}{(1 + s\,\tau_{p2})(1 + s\,\tau_{p3})} \right]. \tag{2.12}$$

The small-signal output voltage ΔV_{out} can also be written as

$$\Delta V_{out} = \frac{A_{dc}}{1 + s\,\tau_0} \cdot \left[F(s) \cdot \Delta V_{in+} - G(s) \cdot \Delta V_{in-} \right], \tag{2.13}$$

where

$$F(s) = \frac{1}{1 + s\,\tau_{p4}} - \frac{1}{2}\,\frac{1}{1 + s\,\tau_{p1}} \cdot \left[\frac{1}{1 + s\,\tau_{p4}} - \frac{1}{(1 + s\,\tau_{p2})(1 + s\,\tau_{p3})} \right], \tag{2.14}$$

$$G(s) = \frac{1}{(1 + s\,\tau_{p2})(1 + s\,\tau_{p3})} + \frac{1}{2}\,\frac{1}{1 + s\,\tau_{p1}} \cdot \left[\frac{1}{1 + s\,\tau_{p4}} - \frac{1}{(1 + s\,\tau_{p2})(1 + s\,\tau_{p3})} \right]. \tag{2.15}$$

From (2.13) we see that the time constant τ_0 introduces a $-90°$ phase shift. Hence if the desired phase margin is at least $60°$, the additional phase shift introduced by the smaller time constants τ_{p1}, τ_{p2}, τ_{p3} and τ_{p4} should not exceed $30°$. In this range, the product $(1 + s\,\tau_{p1})(1 + s\,\tau_{p2})(1 + s\,\tau_{p3})(1 + s\,\tau_{p4})$ can be expanded and approximated by neglecting all the order terms higher than the first order term in s

$$\prod_{k=1}^{4} (1 + s\,\tau_{pk}) \cong 1 + s \sum_{k=1}^{4} \tau_{pk}. \tag{2.16}$$

The above approximation gives a good estimation of the phase shift ϕ for $\phi < 30°$, which correspond to the following frequency range

$$\omega \leq \left(2 \sum_{k=1}^{4} \tau_{pk} \right)^{-1}. \tag{2.17}$$

Using the above approximation leads to

$$F(s) \cong G(s) \cong \frac{1 + s\,\tau_n}{1 + s\,\tau_d}, \tag{2.18}$$

where

$$\tau_n = \tau_{p1} + \frac{\tau_{p2} + \tau_{p3} + \tau_{p4}}{2}, \tag{2.19}$$

$$\tau_d = \tau_{p1} + \tau_{p2} + \tau_{p3} + \tau_{p4}. \tag{2.20}$$

For $\omega \leq 1/(2\tau_d)$, the above expression can be further simplified considering that

$$\frac{1 + s\,\tau_n}{1 + s\,\tau_d} \cong \frac{1}{1 + s\,(\tau_d - \tau_n)} \tag{2.21}$$

where

$$\tau_d - \tau_n = \frac{\tau_{p2} + \tau_{p3} + \tau_{p4}}{2} = \frac{\tau_p}{2}. \tag{2.22}$$

© C. Enz under CC BY-NC-SA

$F(s)$ and $G(s)$ become

$$F(s) \cong G(s) \cong \frac{1}{1 + s\,\frac{\tau_p}{2}}, \tag{2.23}$$

which is valid for $\omega \leq 1/(2\tau_d) = 1/(2(\tau_{p2} + \tau_p))$.

The simplified differential transfer function is then given by

$$A_{dm}(s) \cong \frac{A_{dc}}{(1 + s\,\tau_0)\,(1 + s\,\frac{\tau_p}{2})} = \frac{A_{dc}}{\left(1 + \frac{s}{\omega_0}\right)\left(1 + \frac{s}{\omega_p}\right)} \tag{2.24}$$

with $\omega_0 \triangleq 1/\tau_0$ and $\omega_p \triangleq 2/\tau_p$ where it is assumed that $\tau_p = \tau_{p2} + \tau_{p3} + \tau_{p4} \ll \tau_0$. $\omega_0 \triangleq 1/\tau_0 \ll \omega_p$ is the dominant pole which is set by the load capacitance at the OTA output, whereas $\omega_p \triangleq 2/\tau_p$ represents the non-dominant pole related to all the time constants due to the parasitic capacitances at the current mirror nodes.

The transfer function can be further simplified as

$$A_{dm}(s) \cong \frac{1}{s\,\tau_u\,(1 + s\,\frac{\tau_p}{2})} = \frac{1}{\frac{s}{\omega_u}\left(1 + \frac{s}{\omega_p}\right)} \tag{2.25}$$

where $\omega_u = 1/\tau_u = A_{dc} \cdot \omega_0 = G_m/C_{out}$ is the unity gain frequency or the gain-bandwidth product GBW which is set by the OTA transconductance G_m and the load capacitance C_{out}.

2.2 Noise analysis

We have seen that the noise of the cascode transistors M_4 and M_7 can actually be neglected. Remembering that $A_3 = A_1 \cdot A_2$, the PSD of the output noise current is given by

$$S_{I_{nout}} = A_3^2\,(S_{I_{n1a}} + S_{I_{n1b}} + S_{I_{n2a}} + S_{I_{n2b}}) + A_2^2\,(S_{I_{n2c}} + S_{I_{n3a}}) + S_{I_{n2d}} + S_{I_{n3b}} \tag{2.26}$$

or if we express the output PSD in terms of the output noise conductance

$$S_{I_{nout}} = 4kT \cdot G_{nout} \tag{2.27}$$

where

$$G_{nout} = 2A_3^2 \cdot (G_{n1} + G_{n2}) + A_2^2 \cdot (G_{n2c} + G_{n3a}) + G_{n2d} + G_{n3b} \tag{2.28}$$

where we have assumed that M_{1a} and M_{1b} and M_{2a} and M_{2b} are identical. The G_{ni} are given by

$$G_{ni} = \gamma_{ni} \cdot G_{mi} + G_{mi}^2 \cdot \frac{\rho_i}{W_i\,L_i\,f} \qquad \text{for all transistors.} \tag{2.29}$$

The input-referred noise resistance R_{nin} is then given by

$$R_{nin} \triangleq \frac{G_{nout}}{A_3^2 \cdot G_{m1}^2} = \frac{2(G_{n1} + G_{n2})}{G_{m1}^2} + \frac{G_{n2c} + G_{n3a}}{A_1^2 \cdot G_{m1}^2} + \frac{G_{n2d} + G_{n3b}}{A_3^2 \cdot G_{m1}^2} \tag{2.30}$$

where we have used $A_3 = A_1 \cdot A_2$. As expected, we see that the noise contribution to the input-referred noise of M_{2d} and M_{3b} is divided by A_3^2 and that from M_{2c} and M_{3a} is divided by A_1^2.

We now look at the input-referred thermal and flicker noise separately.

© C. Enz under CC BY-NC-SA

2.2.1 Input-referred thermal noise

The input-referred thermal noise resistance is given by

$$R_{nth} = \frac{2\gamma_{n1}}{G_{m1}} \cdot (1 + \eta_{th}) \tag{2.31}$$

where

$$\eta_{th} = \frac{\gamma_{n2}}{\gamma_{n1}} \frac{G_{m2}}{G_{m1}} + \frac{1}{2A_1^2} \left(\frac{\gamma_{n2c}}{\gamma_{n1}} \frac{G_{m2c}}{G_{m1}} + \frac{\gamma_{n3a}}{\gamma_{n1}} \frac{G_{m3a}}{G_{m1}} \right) \tag{2.32}$$

$$+ \frac{1}{2A_3^2} \left(\frac{\gamma_{n2d}}{\gamma_{n1}} \frac{G_{m2d}}{G_{m1}} + \frac{\gamma_{n3b}}{\gamma_{n1}} \frac{G_{m3b}}{G_{m1}} \right) \tag{2.33}$$

represents the contributions to the input-referred thermal noise of the current mirrors relative to that of the differential pair. Now $G_{m2c} = A_1 \cdot G_{m2a} = A_1 \cdot G_{m2}$, $G_{m3b} = A_2 \cdot G_{m3a} = A_2 \cdot G_{m3}$ and $G_{m2d} = A_3 \cdot G_{m2b} = A_3 \cdot G_{m2}$, and assuming that $\gamma_{n2a} = \gamma_{n2b} \cong \gamma_{n2c} \cong \gamma_{n2d} = \gamma_{n2}$ and $\gamma_{n3a} \cong \gamma_{n3b} = \gamma_{n3}$, then η_{th} reduces to

$$\eta_{th} = \frac{\gamma_{n2}}{2\gamma_{n1}} \frac{G_{m2}}{G_{m1}} \left(2 + \frac{1}{A_1} + \frac{1}{A_3} \right) + \frac{1}{2A_1^2} \frac{\gamma_{n3}}{\gamma_{n1}} \frac{G_{m3}}{G_{m1}} \left(1 + \frac{1}{A_2} \right) \tag{2.34}$$

In the case of unity current gains $A_1 = A_2 = A_3 = 1$, η_{th} simplifies further to

$$\eta_{th} = 2\frac{\gamma_{n2}}{\gamma_{n1}} \frac{G_{m2}}{G_{m1}} + \frac{\gamma_{n3}}{\gamma_{n1}} \frac{G_{m3}}{G_{m1}} \tag{2.35}$$

Of course in the design process, we will try to minimize η_{th} so that the dominant contribution to the input-referred thermal noise comes from the differential pair M_{1a}-M_{1b}.

We can also introduce the OTA thermal noise excess factor as

$$\gamma_{ota} \triangleq G_m \cdot R_{nth}, \tag{2.36}$$

where $G_m = A_3 \cdot G_{m1}$ is the OTA transconductance. This results in

$$\gamma_{ota} = 2A_3 \cdot \gamma_{n1} \cdot (1 + \eta_{th}). \tag{2.37}$$

We see that introducing some current gains larger than one in the current mirrors allows to reduce the contributions of transistors M_{2c}-M_{2d} and M_{3a}-M_{3b}, but on the other hand increases the contribution of the differential pair and hence degrading (i.e. increasing) the OTA thermal noise excess factor γ_{ota}.

For $G_{m1} \gg G_{m2}$ and $G_{m1} \gg G_{m3}$, the noise is dominated by the differential pair and

$$\gamma_{ota} \cong 2A_3 \cdot \gamma_{n1}. \tag{2.38}$$

For unity current gains $A_1 = A_2 = A_3 = 1$, the OTA thermal noise excess factor γ_{ota} reduces to

$$\gamma_{ota} \cong 2\gamma_{n1}. \tag{2.39}$$

2.2.2 Input-referred flicker noise

The input-referred flicker noise resistance is given by

$$\begin{aligned} f \cdot R_{nfl} = & \frac{2\rho_n}{W_{eff1}\, L_{eff1}} + \rho_p \left(\frac{G_{m2}}{G_{m1}} \right)^2 \left(\frac{2}{W_{eff2}\, L_{eff2}} + \frac{1}{W_{eff2c}\, L_{eff2c}} + \frac{1}{W_{eff2d}\, L_{eff2d}} \right) \\ & + \frac{\rho_n}{A_1^2} \left(\frac{G_{m3}}{G_{m1}} \right)^2 \left(\frac{1}{W_{eff3a}\, L_{eff3a}} + \frac{1}{W_{eff3b}\, L_{eff3b}} \right) \end{aligned} \tag{2.40}$$

© C. Enz under CC BY-NC-SA

which can be written as

$$R_{nfl} = \frac{2\rho_n}{W_{eff1}\, L_{eff1} f} \cdot (1 + \eta_{fl}) \tag{2.41}$$

where η_{fl} represents the contributions to the input-referred flicker noise resistance of the current mirrors relative to that of the differential pair

$$\begin{aligned}\eta_{fl} =& \frac{\rho_p}{2\rho_n} \left(\frac{G_{m2}}{G_{m1}}\right)^2 \left(\frac{2W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \frac{W_{eff1}\, L_{eff1}}{W_{eff2c}\, L_{eff2c}} + \frac{W_{eff1}\, L_{eff1}}{W_{eff2d}\, L_{eff2d}}\right) \\ &+ \frac{1}{2A_1^2} \left(\frac{G_{m3}}{G_{m1}}\right)^2 \left(\frac{W_{eff1}\, L_{eff1}}{W_{eff3a}\, L_{eff3a}} + \frac{W_{eff1}\, L_{eff1}}{W_{eff3b}\, L_{eff3b}}\right).\end{aligned} \tag{2.42}$$

For $A_1 = A_2 = A_3 = 1$, $W_{2a} = W_{2b} = W_{2c} = W_{2d} = W_2$, $L_{2a} = L_{2b} = L_{2c} = L_{2d} = L_2$, $W_{3a} = W_{3b} = W_3$ and $L_{3a} = L_{3b} = L_3$ reduces to

$$\eta_{fl} = 2\frac{\rho_p}{\rho_n} \left(\frac{G_{m2}}{G_{m1}}\right)^2 \frac{W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \left(\frac{G_{m3}}{G_{m1}}\right)^2 \frac{W_{eff1}\, L_{eff1}}{W_{eff3}\, L_{eff3}}. \tag{2.43}$$

Similarly to thermal noise, in the design process we will try to minimize η_{fl} in order to reduce the input-referred flicker noise to the contribution of the differential pair M_{1a}-M_{1b}.

The corner frequency is obtained by equating $R_{nfl(f_k)} = R_{nth}$ resulting in

$$f_k = \frac{\rho_n\, G_{m1}}{W_{eff1}\, L_{eff1}\, \gamma_{n1}} \cdot \frac{1 + \eta_{fl}}{1 + \eta_{th}}. \tag{2.44}$$

2.3 Input-referred offset voltage

The input-referred random offset voltage can be obtained in a similar way than noise. The variance of the input-referred offset voltage is given by

$$\sigma^2_{V_{os}} = \sigma^2_{V_{T1}} \cdot (1 + \xi_{V_T}) + \left(\frac{I_b}{G_{m1}}\right)^2 \cdot \sigma^2_{\beta_1} \cdot (1 + \xi_\beta) \tag{2.45}$$

where ξ_{V_T} represents the V_T-mismatch contributions to the input-referred offset of the current mirrors relative to that of the differential pair

$$\xi_{V_T} = 2\left(\frac{G_{m2}}{G_{m1}}\right)^2 \cdot \frac{\sigma^2_{V_{T2}}}{\sigma^2_{V_{T1}}} + \left(\frac{G_{m3}}{A_1\, G_{m1}}\right)^2 \cdot \frac{\sigma^2_{V_{T3}}}{\sigma^2_{V_{T1}}} \tag{2.46}$$

and ξ_β represents the β-mismatch contributions to the input-referred offset of the current mirrors relative to that of the differential pair

$$\xi_\beta = 2\frac{\sigma^2_{\beta_2}}{\sigma^2_{\beta_1}} + \frac{\sigma^2_{\beta_3}}{A_1^2\, \sigma^2_{\beta_1}} \tag{2.47}$$

with

$$\sigma^2_{V_{T1}} = \frac{A^2_{VTn}}{W_{eff1}\, L_{eff1}}, \tag{2.48}$$

$$\sigma^2_{V_{T2}} = \frac{A^2_{VTp}}{W_{eff2}\, L_{eff2}}, \tag{2.49}$$

$$\sigma^2_{V_{T3}} = \frac{A^2_{VTn}}{W_{eff3}\, L_{eff3}}, \tag{2.50}$$

© C. Enz under CC BY-NC-SA

and

$$\sigma_{\beta_1}^2 = \frac{A_{\beta n}^2}{W_{eff1}\, L_{eff1}}, \tag{2.51}$$

$$\sigma_{\beta_2}^2 = \frac{A_{\beta p}^2}{W_{eff2}\, L_{eff2}}, \tag{2.52}$$

$$\sigma_{\beta_3}^2 = \frac{A_{\beta n}^2}{W_{eff3}\, L_{eff3}}. \tag{2.53}$$

Replacing in (2.46) and (2.47) results in

$$\xi_{V_T} = 2\left(\frac{G_{m2}}{G_{m1}}\right)^2 \cdot \left(\frac{A_{VTp}}{A_{VTn}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \left(\frac{G_{m3}}{A_1\, G_{m1}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff3}\, L_{eff3}} \tag{2.54}$$

and

$$\xi_{\beta} = 2\left(\frac{A_{\beta p}}{A_{\beta n}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \frac{1}{A_1^2}\frac{W_{eff1}\, L_{eff1}}{W_{eff3}\, L_{eff3}} \tag{2.55}$$

Similarly to the flicker noise, the input-referred offset (variance or standard deviation) can be reduced by increasing the differential pair $\mathrm{M_{1a}}$-$\mathrm{M_{1b}}$ area $W_{eff1}\, L_{eff1}$ but at the same time also increasing the area of the current mirrors $W_{eff2}\, L_{eff2}$ (and hence $W_{2c}L_{2c}$ and $W_{2d}L_{2d}$) and depending on A_1 also the area of the current mirror $\mathrm{M_{3a}}$-$\mathrm{M_{3b}}$ $W_{3a}L_{3a}$ and $W_{3b}L_{3b}$. This obviously comes at the cost of increased parasitic capacitances at nodes 2, 3 and 4 that will unavoidably reduce the phase margin.

Having done the small-signal, noise and offset analysis, we can now proceed with the design.

© C. Enz under CC BY-NC-SA

3 Design

3.1 Specifications

The OTA specifications are given in Table 3.1.

> **Note**
>
> The specifications given in Table 3.1 are simplified specifications. They are mainly targeting the achievement of a certain gain-bandwidth product GBW and DC gain at lowest current consumption. The GBW sets the differential pair transconductance while the DC gain sets the output conductance. There is an additional specification on the random input-referred offset voltage which, if not met, might eventually require to increase the transistors area. There are no specifications on thermal noise since the transconductance is set by the GBW. There are no specifications on the flicker noise but if the corner frequency was set lower this would required to increase the transistors area. There are also no specifications on the slew-rate, which might be small because of the low-power objective. Finally, there are many more specifications such as CMRR, PSRR, input common-mode voltage range, output-voltage swing, etc... that are not discussed in this example.

Table 3.1: OTA specifications.

Specification	Symbol	Value	Unit
Minimum DC gain	A_{dc}	100	dB
Minimum gain-bandwidth product	GBW	1	MHz
Load capacitance	C_L	1	pF
Maximum input-referred random offset voltage	V_{os}	10	mV
Phase margin	PM	60	$^\circ$

3.2 Process

We will design the symmetrical cascode OTA for a generic 180nm bulk CMOS process. The physical parameters are given in Table 3.2, the global process parameters in Table 3.3 and finally the MOSFET parameters in Table 3.4.

Table 3.2: Physical parameters

Parameter	Value	Unit
T	300	K
U_T	25.875	mV

© C. Enz under CC BY-NC-SA

Table 3.3: Global process parameters

Parameter	Value	Unit
V_{DD}	1.8	V
C_{ox}	8.46	$\frac{fF}{\mu m^2}$
W_{min}	200	nm
L_{min}	180	nm

Table 3.4: Transistor process parameters

Parameter	NMOS	PMOS	Unit
sEKV parameters			
n	1.27	1.31	-
$I_{spec\square}$	715	173	nA
V_{T0}	0.455	0.457	V
L_{sat}	39	12	nm
λ	20	20	$\frac{V}{\mu m}$
Overlap capacitances parameters			
C_{GDo}	0.366	0.328	$\frac{fF}{\mu m}$
C_{GSo}	0.366	0.328	$\frac{fF}{\mu m}$
C_{GBo}	0	0	$\frac{fF}{\mu m}$
Junction capacitances parameters			
C_J	1	1.121	$\frac{fF}{\mu m^2}$
C_{JSW}	0.204	0.248	$\frac{fF}{\mu m}$
Flicker noise parameters			
K_F	8.1e-24	6.8e-23	J
AF	1	1	-
ρ	0.05782	0.4818	$\frac{V \cdot m^2}{A \cdot s}$
Matching parameters			
A_{VT}	5	5	$mV \cdot \mu m$
A_β	1	1	$\% \cdot \mu m$
Source and drain sheet resistance parameter			
R_{sh}	600	2386	$\frac{\Omega}{\mu m}$
Width and length parameters			
ΔW	39	54	nm
ΔL	-76	-72	nm

Warning

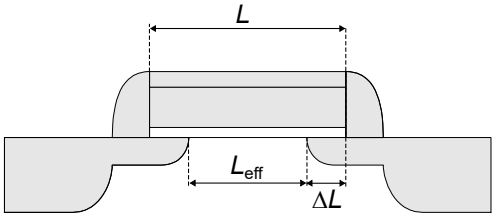


Figure 3.1: Definition of the effective length.

© C. Enz under CC BY-NC-SA

In Spice, the symbols W and L correspond to the *drawn* width and length. As shown in Figure 3.1, the *effective* width and length are then defined as

$$W_{eff} = W + \Delta W, \tag{3.1}$$
$$L_{eff} = L + \Delta L, \tag{3.2}$$

where usually ΔL has a negative sign as shown in Table 3.4. However, ΔW may be positive or negative. In the special case of this generic 180 nm CMOS process, it is positive as shown in Table 3.4.

3.3 Design procedure

Important

For this process, the transistor dimensions are rounded to 10nm.

We need to start with choosing the curent gain A_3. We have seen that having a current gain $A_3 > 1$ improves the transconductance but at the cost of a higher thermal noise excess factor γ_{ota}. We therefore will set to $A_1 = 1$, $A_2 = 1$ and $A_3 = 1$. Since $A_1 = A_2 = A_3 = 1$, transistors M_{2a}, M_{2b}, M_{2c} and M_{2d} are all identical and so are M_{3a} and M_{3b}. We therefore have the following variables:

- M_{1a}-M_{1b}: W_1, L_1 and I_b. Note that the current I_b is set by G_{m1} because M_{1a}-M_{1b} are biased in weak inversion,
- M_{2a}-M_{2c} and M_{2b}-M_{2d}: IC_2, W_2 and L_2 since the current is already set,
- M_{3a}-M_{3b}: IC_3, W_3 and L_3 since the current is already set,
- M_4: IC_4, W_4 and L_4 since the current is already set,
- M_7: IC_7, W_7 and L_7 since the current is already set,
- M_{5a}-M_{5b}: IC_5, W_5 and L_5 since the current is already set.

We start sizing the differential pair M_{1a}-M_{1b}.

3.3.1 Sizing the differential pair M_{1a}-M_{1b}

Code

The code used for the design is detailed below.

```
# Sizing of M1a-M1b
Ibmin=2*pi*n0n*UT*CL*GBWspec
SRmin=Ibmin/CL
Ib=250E-9
IC1=0.1
Gm1=Ib/(n0n*UT)*gmsid_ic(IC1)
GBW=Gm1/(2*pi*CL)
WoverL1=Ib/(Ispecsqn*IC1)
Go=Gm1/Adc
Leff1=2*Ib/(lambdan*Go)
L1=Leff1-DLn
L1=round(1E8*L1)/1E8
Weff1=WoverL1*Leff1
W1=Weff1-DWn
```

```
W1=round(1E8*W1)/1E8
```

In this example there is no specification on the input-referred white noise. Therefore the transconductance G_{m1} is set by the gain-bandwidth product according to

$$GBW = \frac{G_{m1}}{2\pi C_{out}}, \tag{3.3}$$

where G_{m1} is the gate transconductance of M_{1a} and M_{1b} and C_{out} the total output capacitance

$$C_{out} = C_L + C_o \tag{3.4}$$

with C_L the specified load capacitance and C_o the parasitic capacitance at the output node.

In order to minimize the input-referred noise and offset, the input differential pair should be biased in weak inversion. The transconductance G_{m1} in deep weak inversion is then given by

$$G_{m1} = \frac{I_b}{n\,U_T}. \tag{3.5}$$

The bias current I_b is the current flowing in each transistor M_{1a} and M_{1b} when the input differential voltage is zero. The bias current provided by M_{5b} is therefore $2I_b$. The bias current must satisfy the following inequality

$$I_b \geq 2\pi\, n_{0n}\, U_T\, C_L\, GBW_{min}. \tag{3.6}$$

which for the given specifications gives $I_{b,min} = 207\ nA$. The corresponding slew-rate is then equal to $SR_{min} = 207\ mV/\mu s$ which we will consider as sufficient.

> **Important**
>
> If the slew-rate is not sufficient, the bias current I_b should be increased resulting in a higher current and power consumption. Other options include the use of a class AB OTA [11] or a dynamic/adaptive biasing OTA [12].

As we will see below, the cascode transistors M_4 and M_7 will be biased in weak inversion to save voltage headroom at the output and preserve the output voltage swing of the symmetrical OTA. They will therefore be rather large transistors. Since they are connected to the output, they will increase the parasitic capacitance at the output C_o due mostly to the junction capacitances. To have sufficient margin, we set $I_b = 250\ nA$ and the inversion coefficient to $IC_1 = 0.1$. The transconductance can be calculated from the G_m/I_D function as $G_{m1} = 6.962\ \mu A/V$. Neglecting C_o, this leads to a gain-bandwidth product $GBW = 1.1\ MHz$, which is slightly higher than the target specification offering some margin to account for the additionnal parasitic capacitance C_o. Knowing the drain current I_{D1} and the inversion coefficient, we can calculate the aspect ratio $W_1/L_1 = 3.5$.

Before finalizing the sizing of the differential pair, we first will size the current mirrors.

3.3.2 Sizing the pMOS current mirrors M_{2a}-M_{2c} and M_{2b}-M_{2d}

> **Code**
>
> The code used for the design is detailed below.

```
Vicmax=1.3
VGS1=VT0n+n0n*UT*vps_ic(IC1)
VDSsat1=UT*vdssat_ic(IC1)
VSG2=VDD-Vicmax+VGS1-VDSsat1
IC2=ic_vps((VSG2-VT0p)/(n0p*UT))
```

© C. Enz under CC BY-NC-SA

The gate voltage of M_{2a}-M_{2c} and M_{2a}-M_{2b} should be set as low as possible for a given maximum common mode input voltage $V_{ic,max}$ still keeping M_{1a}-M_{1b} in saturation. The source-to-gate voltage of M_{2a}-M_{2b} V_{SG2} is given by

$$V_{SG2} = V_{DD} - V_{ic,max} + V_{GS1} - V_{DSsat1}. \tag{3.7}$$

The gate-to-source voltage V_{GS1} and the saturation voltage V_{DSsat1} of M_{1a}-M_{1b} are given by $V_{GS1} = 382\ mV$ and $V_{DSsat1} = 105\ mV$. For a maximum input common-mode voltage $V_{ic,max} = 1.3\ V$, the minimum source-to-gate voltage is given by $V_{SG2} = 778\ mV$, corresponding to an inversion coefficient $IC_2 = 20.382$.

Code

The code used for the design is detailed below.

```
VSG2=0.8
IC2=ic_vps((VSG2-VT0p)/(n0p*UT))
VDSsat2=UT*vdssat_ic(IC2)
Ispec2=Ib/IC2
Gm2=Ispec2/(n0p*UT)*gms_ic(IC2)
WoverL2=Ispec2/Ispecsqp
```

We can set V_{SG2} to 800 mV which corresponds to an inversion coefficient $IC_2 = 23.158$ and a saturation voltage $V_{DSsat2} = 270\ mV$. The specific current I_{spec}, aspect ratio and transconductance are then given by $I_{spec2} = 10.80\ nA$, $W_2/L_2 = 0.062$ and $G_{m2} = 1.386\ \mu A/V$.

Code

The code used for the design is detailed below.

```
# Sizing of M2a-M2c M2b-M2d
fp2GBW=10
fp2=fp2GBW*GBW
wp2=2*pi*fp2
C2=Gm2/wp2
CWL=2*Cox*(cgsi_ic(IC2)+cgbi_ic(IC2,n0p))
CW=2*(CGSOp+CGBOp)
Weff2=(-CW*WoverL2+sqrt(WoverL2)*sqrt(4*C2*CWL+CW**2*WoverL2))/(2*CWL)
#L2=(-CW*WoverL2+sqrt(WoverL2)*sqrt(4*C2*CWL+CW**2*WoverL2))/(2*CWL*WoverL2)
W2=Weff2-DWp
W2=round(1E8*W2)/1E8
Leff2=Weff2/WoverL2
L2=Leff2-DLp
L2=round(1E8*L2)/1E8
```

We now have to make sure that the non-dominant poles f_{p2} and f_{p4} at nodes 2 and 4 are sufficiently higher than the GBW to insure the desired phase margin. The non-dominant pole is given by

$$\omega_{p2} = \frac{G_{m2}}{C_2}, \tag{3.8}$$

where C_2 is the total capacitance at node 2

$$C_2 = 2(C_{GS2} + C_{GB2}). \tag{3.9}$$

Assuming M_{2a}-M_{2b} are in saturation, the gate-to-source capacitance C_{GS2} is given by

$$C_{GS2} \cong W_{eff2}\, L_{eff2}\, C_{ox} \cdot c_{gsi} + C_{GSop} \cdot W_{eff2}, \tag{3.10}$$

© C. Enz under CC BY-NC-SA

where C_{GSop} is the gate-to-source overlap capacitance per unit width for pMOS transistors. c_{gsi} is the intrinsic gate-to-source capacitance normalized to the total gate capacitance $W_{eff}\,L_{eff}\,C_{ox}$, which is typically equal to 2/3 in strong inversion and is proportionnal to IC in weak inversion.

The gate-to-bulk capacitance C_{GB2} is given by

$$C_{GB2} \cong W_{eff2}\,L_{eff2}\,C_{ox}\cdot c_{gbi} + C_{GBop}\cdot W_{eff2}, \tag{3.11}$$

where C_{GBop} is the gate-to-bulk overlap capacitance per unit width and c_{gbi} is the gate-to-bulk intrinsic capacitance normalized to the total gate capacitance $W_{eff}\,L_{eff}\,C_{ox}$ and given by

$$c_{gbi} = \frac{n-1}{n}\cdot c_{gsi}. \tag{3.12}$$

The capacitance at node 2 then scales with W_{eff2} and L_{eff2} according to

$$C_2 = W_{eff2}\,L_{eff2}\cdot C_{WL} + W_{eff2}\cdot C_W, \tag{3.13}$$

with

$$C_{WL} = 2\,C_{ox}\cdot(c_{gsi}+c_{gbi}), \tag{3.14}$$
$$C_W = 2(C_{GSop}+G_{GBop}). \tag{3.15}$$

We already have set the inversion coefficient $IC_2 = 23.158$, from which we can calculate $c_{gsi} = 0.593$ and $c_{gbi} = 0.095$. Since the aspect ratio $AR_2 \triangleq W_{eff2}/L_{eff2}$ has already been set by the transconductance and the current, we can derive W_{eff2} and L_{eff2} for achieving a given capacitance C_2 according to

$$W_{eff2} = \frac{-C_W\cdot AR_2 + \sqrt{AR_2}\cdot\sqrt{4\,C_2\,C_{WL}+C_W^2\cdot AR_2}}{2\,C_{WL}}, \tag{3.16}$$

$$L_{eff2} = \frac{W_{eff2}}{AR_2}. \tag{3.17}$$

Setting the non-dominant pole f_{p2} to 10 times the GBW, we get $C_2 = 20$ fF, $L_{eff2} = 5.21\ \mu m$, $L_2 = 5.28\ \mu m$ and $W_{eff2} = 325\ nm$, $W_2 = 270\ nm$.

Since $A_1 = A_3 = 1$, M_{2a}, M_{2b}, M_{2c} and M_{2d} are all identical and hence $W_{2a} = W_{2b} = W_{2c} = W_{2d} = 270\ nm$ and $L_{2a} = L_{2b} = L_{2c} = L_{2d} = 5.28\ \mu m$.

We next size the nMOS current mirror M_{3a} and M_{3b}.

3.3.3 Sizing the nMOS current mirror M_{3a} and M_{3b}

To size M_{3a} and M_{3b} we need to consider the output swing and make sure that M_{2c} remains in saturation for the chosen overdrive voltage chosen for M_{3a}.

Code

The code used for the design is detailed below.

```
# Sizing of M3a-M3b
Vosw=1
Vomin=(VDD-Vosw)/2
IC7=0.1
VDSsat7=UT*vdssat_ic(IC7)
VDSsat3=Vomin-VDSsat7
IC3=ic_vdssat(VDSsat3/UT)
```

```
VGS3a=VT0n+n0n*UT*vps_ic(IC3)
VSD2c=VDD-VGS3a
VDSsat2c=UT*vdssat_ic(IC2)
Ispec3=Ib/IC3
Gm3=Ispec3/(n0p*UT)*gms_ic(IC3)
WoverL3=Ispec3/Ispecsqn
```

Let's start maximizing the output swing assuming that we bias the cascode transistor M_7 in weak inversion for minimum saturation voltage and maximum G_m/I_D. We set the inversion coefficient of M_7 to $IC_7 = 0.1$. The saturation voltage of M_7 is then given by $V_{DSsat7} = 105\ mV$.

Let's say that we want an output swing of 1 V. This means that we have 800 mV to share between the positive and negative saturation voltages which we decide to split equally. This means 400 mV for the saturation voltages of M_7 and M_{3b} which leaves a saturation voltage for M_{3b} of $V_{DSsat3} = 295\ mV$. This corresponds to an inversion coefficient $IC_3 = 28.5$.

We now can check whether with the chosen inversion coefficient for M_{3b}, M_{2c} remains in saturation. The gate-to-source voltage of M_{3a} is $V_{GS3a} = 827\ mV$, giving a source to-drain voltage for M_{2c} $V_{SD2c} = 973\ mV$, which is larger than its saturation voltage $V_{DSsat2c} = 270\ mV$.

Knowing the current and the inversion coefficient $IC_3 = 28.5$, we can deduce $I_{spec3} = 8.759\ nA$, $W_3/L_3 =$ 1.2e-02 and $G_{m3} = 1.261\ \mu A/V$.

Code

The code used for the design is detailed below.

```
# Sizing of M3a-M3b
fp3GBW=10
fp3=fp3GBW*GBW
wp3=2*pi*fp3
C3=Gm3/wp3
CWL3=2*Cox*(cgsi_ic(IC3)+cgbi_ic(IC3,n0n))
CW3=2*(CGSOn+CGBOn)
Weff3=(-CW3*WoverL3+sqrt(WoverL3)*sqrt(4*C3*CWL3+CW3**2*WoverL3))/(2*CWL3)
W3=Weff3-DWn
W3=round(1E8*W3)/1E8
Leff3=Weff3/WoverL3
L3=Leff3-DLn
L3=round(1E8*L3)/1E8
```

Similarly to what was done for transistor M_{2a}-M_{2c}, we need to make sure that the non-dominant pole f_{p3} at node 3 remains much higher than the GBW. Setting $f_{p3}/GBW = 10$ and proceeding in a similar way we get $f_{p3}/GBW = 10$ with $C_3 = 18.1\ fF$ resulting in $W_3 = 100\ nm$ and $L_3 = 11.34\ \mu m$.

We see that W_3 is smaller than the minimum width. If we don't want to increase $W_3\,L_3$ and hence C_3, we need to reduce the inversion coefficient IC_3. We can find the IC such that $W_3 = W_{min}$ for the given f_{p3} by looking at Figure 3.2.

Code

The code used for the design is detailed below.

```
# Sizing of M3a-M3b

from scipy.optimize import fsolve
```

© C. Enz under CC BY-NC-SA

```
def fic(IC):
    Ispec=Ib/IC
    Gm=Ispec/(n0n*UT)*gms_ic(IC)
    WoverL=Ispec/Ispecsqn
    C3=Gm/wp3
    CWL=2*Cox*(cgsi_ic(IC)+cgbi_ic(IC,n0n))
    Weff=(-CW*WoverL+sqrt(WoverL)*sqrt(4*C3*CWL+CW**2*WoverL))/(2*CWL)
    W=Weff-DWn
#    print(W)
    return W-Wmin

fp3=fp3GBW*GBW
wp3=2*pi*fp3
CW=2*(CGSOn+CGBOn)
ICguess=20
ICopt=fsolve(fic,ICguess)
IC3ini=IC3
IC3=ICopt[0]
#print(f'ICopt = {IC3:.1f}')
Ispec3=Ib/IC3
Gm3=Ispec3/(n0n*UT)*gms_ic(IC3)
WoverL3=Ispec3/Ispecsqn
#W3=(-CWn*WoverL3+sqrt(WoverL3)*sqrt(4*C3*CWLn+CWn**2*WoverL3))/(2*CWLn)
W3=Wmin
Weff3=W3+DWn
Leff3=Weff3/WoverL3
L3=Leff3-DLn
CWL=2*Cox*(cgsi_ic(IC3)+cgbi_ic(IC3,n0n))
C3=W3*Leff3*CWL+Weff3*CW
wp3=Gm3/C3
fp3=wp3/(2*pi)
```

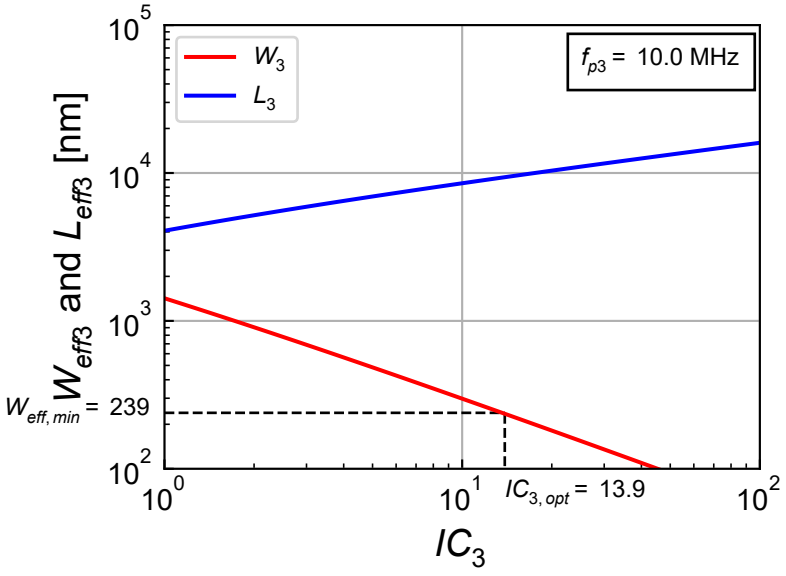


Figure 3.2: Length L_3 and width W_3 of M_3 versus IC_3 for a given f_{p3}/GBW ratio.

We get $IC_3 = 13.9$, $W_3 = 200\ nm$ and $L_3 = 9.58\ \mu m$ with $C_3 = 11.2\ fF$ and $f_{p3}/GBW = 10$.

The inversion coefficient of M_{3a}-M_{3b} is therefore reduced from 28.5 to 13.9 in order to maintain a reasonable parasitic capacitance at node 3 to make sure that f_{p3} is 10-times higher than the GBW.

M_{3b} is identical to M_{3a} and therefore $W_{3a} = W_{3b} = 200\ nm$ and $L_{3a} = L_{3b} = 9.58\ \mu m$.

3.3.4 Sizing the cascode transistors M_4-M_7

Code

The code used for the design is detailed below.

```
# Sizing of M7
Go=A3*Gm1/Adc
Ispec7=Ib/IC7
WoverL7=Ispec7/Ispecsqn
Gms7=Ib/UT*gmsid_ic(IC7)
Gds3=Ib/(lambdan*Leff3)
Gds7=Gms7/Gds3*Go/2
Leff7=Ib/(lambdan*Gds7)
L7=Leff7-DLn
L7=round(1E8*L7)/1E8
Weff7=WoverL7*Leff7
W7=Weff7-DWn
W7=round(1E8*W7)/1E8
VDSsat3=UT*vdssat_ic(IC3)
VGS7=VT0n
Vb2min=VDSsat3+VGS7
Vb2=VDD/2
```

The cascode transistors are sized according to the desired DC gain given by

$$A_{dc} = \frac{A_3 \cdot G_{m1}}{G_o} \tag{3.18}$$

where

$$G_o = G_{on} + G_{op} \tag{3.19}$$

with

$$G_{on} = \frac{G_{ds3b} \cdot G_{ds7}}{G_{ms7}} \tag{3.20}$$

$$G_{op} = \frac{G_{ds2d} \cdot G_{ds4}}{G_{ms4}} \tag{3.21}$$

the output conductance of of the nMOS and pMOS cascode stages, respectively. The minimum DC gain specification is given by $A_{dc} = 1.0\text{e+}05$ or $A_{dc} = 100$ dB. Since the transconductance G_{m1} is already set by the GBW, the output conductance at the output node G_o is then given by $G_o = G_{m1}/A_{dc} = 0.070\ nA/V$. We will split the output conductance equally between the nMOS and pMOS cascodes. We already have set the inversion coefficient of M_7 to $IC_7 = 0.1$. Knowing the current, we can deduce the specific current $I_{spec7} = 2.5\ \mu A$ and the aspect ratio $W_7/L_7 = 3.5$. Knowing the current and IC_7 we can get $G_{ms7} = 8.851\ \mu A/V$.

We already have chosen the length of M_{3a}-M_{3b} $L_3 = 9.58\ \mu m$. The output conductance of M_{3b} can then be estimated to $G_{ds3} = 1.315\ nA/V$. We can then deduce the output conductance of M_7 as $G_{ds7} = 234.245\ nA/V$ from which we deduce its length $L_7 = 0.13\ \mu m$ which is smaller than the minimum length.

© C. Enz under CC BY-NC-SA

Code

The code used for the design is detailed below.

```
# Sizing of M7
L7=360e-9
Leff7=L7+DLn
Weff7=WoverL7*Leff7
W7=Weff7-DWn
W7=round(1E8*W7)/1E8
```

To have some margin and avoid DIBL, we choose a slightly longer length $L_7 = 360\ nm$ ($L_{eff7} = 284\ nm$). Keeping the same $W_7/L_7 = 3.5$, we get $W_7 = 0.95\ \mu m$ ($W_{eff7} = 0.99\ \mu m$).

The minimum value of the bias voltage V_{b2} is set such that M_{3b} remains in saturation

$$V_{b2,min} = V_{DSsat3} + V_{GS2}. \tag{3.22}$$

The saturation voltage of M_{3b} is equal to $V_{DSsat3} = 219\ mV$ and the V_{GS} voltage of M_7 is $V_{GS7} \cong 455\ mV$, which results $V_{b2,min} = 674\ mV$. We finally choose $V_{b2} = V_{DD}/2 = 900\ mV$.

Code

The code used for the design is detailed below.

```
# Sizing of M4
IC4=IC7
VDSsat4=UT*vdssat_ic(IC4)
Ispec4=Ib/IC4
WoverL4=Ispec4/Ispecsqp
Gms4=Ib/UT*gmsid_ic(IC4)
Gds2=Ib/(lambdap*L2)
Gds4=Gms4/Gds2*Go/2
Leff4=Ib/(lambdap*Gds4)
L4=Leff4-DLp
L4=round(1E8*L4)/1E8
Weff4=WoverL4*Leff4
W4=Weff4-DWp
W4=round(1E8*W4)/1E8
```

For M_4 we set the same inversion coefficient as M_7 $IC_4 = IC_7 = 0.1$, from which we get its saturation voltage $V_{DSsat4} = 105\ mV$. Knowing the current and the IC we can deduce $I_{spec4} = 2.5\ \mu A$ and $W_4/L_4 = 14.4$. The source transconductance of M_4 is then given by $G_{ms4} = 8.851\ \mu A/V$. The output conductance of M_{2d} can be estimated as $G_{ds2} = 2.367\ nA/V$. We can then deduce the output conductance of M_4 as $G_{ds4} = 130.140\ nA/V$, from which we can calculate the length $L_4 = 170\ nm$, which is smaller than the minimum length.

Code

The code used for the design is detailed below.

```
# Sizing of M4
L4=250e-9
Leff4=L4+DLp
Weff4=WoverL4*Leff4
W4=Weff4-DWp
```

© C. Enz under CC BY-NC-SA

```
W4=round(1E8*W4)/1E8
```

We could choose $L_4 = L_{min}$, but in order to have some margin on the DC gain, we choose $L_4 = 250$ nm ($L_{eff4} = 178$ nm). Keeping the same $W_4/L_4 = 14.4$, this leads to $W_4 = 2.52$ μm ($W_{eff4} = 2.57$ μm).

Code

The code used for the design is detailed below.

```
VSDsat2=UT*vdssat_ic(IC2)
VSG4=VT0p
Vb1max=VDD-VSDsat2-VSG4
#Vb1=VDD/2
Vb1=0.7
```

The maximum value of the bias voltage V_{b1} is limited by the saturation voltage of M_{2d} according to

$$V_{b1,max} = V_{DD} - V_{SDsat2d} - V_{SG4}. \tag{3.23}$$

The saturation voltage of M_{2d} is $V_{SDsat2} = 270$ mV and the source-to-gate voltage of M_4 is $V_{SG4} \cong 457$ mV, leading to $V_{b1,max} \cong 1.073$ V. For convenience we could also set $V_{b1} = V_{DD}/2 = 700$ mV. However, in the simulations below, we see that the output conductance of M_4 is too large degrading the DC gain. This can be corrected by increasing the V_{DS} voltage of M_4 therefore reducing its output conductance allowing to reach the desired DC gain.

Code

The code used for the design is detailed below.

```
# Check the resulting DC gain
Leff3=L3+DLn
Gds3b=Ib/(lambdan*Leff3)
Leff7=L7+DLn
Ispec7=Ispecsqn*Weff7/Leff7
IC7=Ib/Ispec7
Gms7=Ib/UT*gmsid_ic(IC7)
Gds7=Ib/(lambdan*L7)
Gon=Gds3b*Gds7/Gms7
Leff2=L2+DLp
Gds2d=Ib/(lambdap*Leff2)
Weff4=W4+DWp
Leff4=L4+DLp
Ispec4=Ispecsqp*Weff4/Leff4
IC4=Ib/Ispec4
Gms4=Ib/UT*gmsid_ic(IC4)
Gds4=Ib/(lambdap*L4)
Gop=Gds2d*Gds4/Gms4
Go=Gon+Gop
Adcest=Gm1/Go
AdcdBest=20*log(Adcest)
```

We can now check the resulting DC gain. The output conductance of the nMOS and pMOS cascodes are $G_{on} = 5.160$ pA/V and $G_{op} = 13.557$ pA/V, respectively. This results in a total output conductance

© C. Enz under CC BY-NC-SA

$G_o = 18.717\ pA/V$ and a DC gain $A_{dc} = 4e{+}05$ or $A_{dc} = 111.4$ dB, which is 11.4 dB higher than the specifications.

3.3.5 Sizing the nMOS current mirror M_{5a}-M_{5b}

Code

The code used for the design is detailed below.

```
# Sizing of M5a-M5b
Vicmin=0.6
VGS1=VT0n+n0n*UT*vps_ic(IC1)
VDSsat5=Vicmin-VGS1
IC5=ic_vdssat(VDSsat5/UT)
Ispec5=2*Ib/IC5
WoverL5=Ispec5/Ispecsqn
W5=Wmin
Weff5=W5+DWn
Leff5=Weff5/WoverL5
L5=Leff5-DLn
L5=round(1E8*L5)/1E8
```

Finally we need to size M_{5a}-M_{5b}. The minimum input common mode voltage $V_{ic,min}$ is limited by the saturation voltage of M_{5b} according to

$$V_{ic,min} = V_{GS1} + V_{DSsat5b}. \tag{3.24}$$

Since M_{1a}-M_{1b} are in a separate well, their gate-to-source voltage is $V_{GS1} = V_{T0n} + n\,U_T\,(v_p - v_s) \cong 382\ mV$. Setting the minimum input common mode voltage to $V_{ic,min} = 600\ mV$ leads to the maximum saturation voltage of M_{5b} $V_{DSsat5} = V_{ic,min} - V_{GS1} \cong 218\ mV$. The corresponding inversion coefficient of M_{5b} is $IC_5 = 13.7$. From I_b and IC_5 we get $I_{spec5} = 36.547\ nA$ and $W_5/L_5 = 5.111e{-}02$. With such a low W/L ratio, we need to set W_5 to the minimum width $W_5 = W_{min} = 200\ nm$ from which we deduce $L_5 = 4.75\ \mu m$.

Transistor M_{5a} is identical to M_{5b} and hence $W_{5a} = W_{5b} = 200\ nm$ ($W_{eff5} = 239\ nm$) and $L_{5a} = L_{5b} = 4.75\ \mu m$ ($L_{eff5} = 4.68\ \mu m$).

The sizing process is almost finished, but we still need to finalize the sizing of the differential pair M_{1a}-M_{1b}.

3.3.6 Final sizing of the differential pair M_{1a}-M_{1b}

Code

The code used for the design is detailed below.

```
# Final sizing of M1a-M1b
flickerfactor=1
WL1=flickerfactor/(rhop/rhon*(Gm2/Gm1)**2*2/(W2*L2)+(Gm3/Gm1)**2/(W3*L3))
Weff1=sqrt(WoverL1*WL1)
W1=Weff1-DWn
W1=round(1E8*W1)/1E8
Leff1=sqrt(WL1/WoverL1)
L1=Leff1-DLn
```

© C. Enz under CC BY-NC-SA

```
L1=round(1E8*L1)/1E8
flickerpMOSCM=rhop/rhon*(Gm2/Gm1)**2*2*Weff1*Leff1/(Weff2*Leff2)
flickernMOSCM=(Gm3/Gm1)**2*Weff1*Leff1/(Weff3*Leff3)
gamman1=gamman_ic(IC1,n0n)
gamman2=gamman_ic(IC2,n0p)
gamman3=gamman_ic(IC3,n0n)
etath=2*gamman2/gamman1*Gm2/Gm1+gamman3/gamman1*Gm3/Gm1
gammaota=2*gamman1*(1+etath)
Rninth=gammaota/Gm1
Sninth=4*kT*Rninth
SninthdBv=10*log(Sninth)
etafl=2*rhop/rhon*(Gm2/Gm1)**2*Weff1*Leff1/(Weff2*Leff2)+(Gm3/Gm1)**2*Weff1*Leff1/(Weff3*Lef
fk=rhon*Gm1/(W1*L1*gamman1)*(1+etafl)/(1+etath)
```

We haven't finalized the sizing of M_{1a}-M_{1b}. We already have the W/L ratio but still need to set W_1 and L_1. We don't have any specifications on the noise and particularly the corner frequency, while the white noise is already set by the transconductance G_{m1}.

Since the current mirrors current gains are all unity $A_1 = A_2 = A_3 = 1$, the transistors M_{2a}, M_{2b}, M_{2c} and M_{2d} are all identical. In this case the contributions of the current mirrors to the input-referred flicker noise relative to the contribution of the differential pair is captured by the η_{fl} parameter recalled below

$$\eta_{fl} = 2\,\frac{\rho_p}{\rho_n}\cdot\left(\frac{G_{m2}}{G_{m1}}\right)^2\,\frac{W_{eff1}\,L_{eff1}}{W_{eff2}\,L_{eff2}} + \left(\frac{G_{m3}}{G_{m1}}\right)^2\cdot\frac{W_{eff1}\,L_{eff1}}{W_{eff3}\,L_{eff3}}. \tag{3.25}$$

The ratio of the flicker noise parameter of pMOS transistors to that of nMOS transistors is given by $\rho_p/\rho_n = 8.333$, which is rather large. The square of the transconductance ratio $(G_{m1}/G_{m2})^2 = 25.237$ should compensate for this large ρ_p/ρ_n ratio. We can try to make the contribution of the input differential pair equal to the contribution of the current mirrors (i.e. set $\eta_{fl} = 1$). This results in $W_1 \cdot L_1 = 2.010\ \mu m^2$.

Since we already have W_1/L_1 we can deduce $W_1 = 2.61\ \mu m$ ($W_{eff1} = 2.65\ \mu m$) and $L_1 = 830\ nm$ ($L_{eff1} = 758\ nm$). Since transistor M_{1a} is identical to M_{1b}, we have $W_{1a} = W_{1b} = 2.61\ \mu m$ and $L_{1a} = L_{1b} = 830\ nm$.

The relative contribution to the input-referred flicker noise of the pMOS current mirrors M_{2a} to M_{2d} is 0.785, whereas that of the nMOS current mirror M_{3a} and M_{3b} is 0.058. This leads to $\eta_{fl} = 0.843$, which is close to the desired $\eta_{fl} \cong 1$.

We now have finalized the design. We can still check the thermal noise. The contributions of the current mirrors to the input-referred white noise relative to that of the differential pair is $\eta_{th} = 0.818$. This results in an OTA thermal noise excess factor $\gamma_{ota} = 2.4$. The input-referred white noise resistance is $R_{ninth} = 341.4\ k\Omega$, corresponding to a PSD $S_{ninth} = 5.7\text{e-}15\ V^2/Hz$ or $\sqrt{S_{ninth}} = 75.2\ nV/\sqrt{Hz}$ or $10 \cdot \log(S_{ninth}) = -142.5\ dBv/\sqrt{Hz}$. This sets the corner frequency to $f_k = 288.2\ kHz$.

The design is now finalized and is summarized in the section.

3.4 Summary

3.4.1 Specifications

The specifications are recalled in Table 3.5.

Table 3.5: OTA specifications.

Specification	Symbol	Value	Unit
Minimum DC gain	A_{dc}	100	dB
Minimum gain-bandwidth product	GBW	1	MHz
Load capacitance	C_L	1	pF
Maximum input-referred random offset voltage	V_{os}	10	mV
Phase margin	PM	60	$^\circ$

3.4.2 Bias

The bias information are summarized in Table 3.6.

Table 3.6: OTA bias.

Bias voltage or current	Symbol	Value	Unit
Supply voltage	V_{DD}	1.8	V
Bias current	I_b	250.0	nA
Cascode bias voltage	V_{b1}	0.7	V
Cascode bias voltage	V_{b2}	0.9	V

3.4.3 Transistor information

The transistor sizes and large-signal variables are summarized in Table 3.7, whereas Table 3.8 gives the small-signal and thermal noise parameters. An Excel table is generated with more information (e.g. all the parasitic capacitances).

Table 3.7: Transistor size and bias information.

Transistor	W $[\mu m]$	L $[\mu m]$	I_D $[nA]$	I_{spec} $[nA]$	IC	$V_G - V_{T0}$ $[mV]$	V_{DSsat} $[mV]$
M1a	2.61	0.83	250	2512	0.1	-45	105
M1b	2.61	0.83	250	2512	0.1	-45	105
M2a	0.27	5.28	250	11	23.2	201	270
M2b	0.27	5.28	250	11	23.2	201	270
M2c	0.27	5.28	250	11	23.2	201	270
M2d	0.27	5.28	250	11	23.2	201	270
M3a	0.20	9.58	250	18	13.9	157	219
M3b	0.20	9.58	250	18	13.9	157	219
M4	2.52	0.25	250	2504	0.1	-44	105
M5a	0.20	4.75	500	37	13.7	155	218
M5b	0.20	4.75	500	37	13.7	155	218
M7	0.95	0.36	250	2490	0.1	-45	105

Table 3.8: Transistor small-signal and thermal noise parameters.

Transistor	G_{spec} $[\mu A/V]$	G_{ms} $[\mu A/V]$	G_m $[\mu A/V]$	G_{ds} $[nA/V]$	γ_n
M1a	97.082	8.854	6.964	15.060	0.653
M1b	97.082	8.854	6.964	15.060	0.653
M2a	0.416	1.808	1.384	2.367	0.830

© C. Enz under CC BY-NC-SA

Table 3.8: Transistor small-signal and thermal noise parameters.

Transistor	G_{spec} $[\mu A/V]$	G_{ms} $[\mu A/V]$	G_m $[\mu A/V]$	G_{ds} $[nA/V]$	γ_n
M2b	0.416	1.808	1.384	2.367	0.830
M2c	0.416	1.808	1.384	2.367	0.830
M2d	0.416	1.808	1.384	2.367	0.830
M3a	0.695	2.267	1.783	1.305	0.798
M3b	0.695	2.267	1.783	1.305	0.798
M4	96.758	8.852	6.778	50.000	0.671
M5a	1.413	4.566	3.592	5.263	0.798
M5b	1.413	4.566	3.592	5.263	0.798
M7	96.229	8.848	6.960	34.722	0.654

© C. Enz under CC BY-NC-SA

4 OTA Characteristics

In this section, we check whether the specs are achieved.

4.1 Open-loop gain

We can calculate the various OTA features related to the open-loop transfer function, which are given in Table 4.1.

Table 4.1: OTA gain variables.

Symbol	Theoretical Value	Unit
A_{dc}	112	dB
G_{m1}	6.964	$\mu A/V$
G_{m2}	1.384	$\mu A/V$
G_{m3}	1.783	$\mu A/V$
f_0	2.932	Hz
GBW	1.104	MHz
f_{p2}	11.096	MHz
f_{p3}	11.08	MHz
f_{p4}	11.096	MHz

The gain-bandwidth product from the specifications is repeated here

$GBW = 1.000$ MHz (from spec).

The estimated value assuming that all the non-dominant poles are much higher than the GBW is given by

$GBW_{est} = 1.104$ MHz (estimation).

The GBW accounting for the effect of the additional non-dominant poles is given by

$GBW_{the} = 1.095$ MHz (theory).

We see that there is only a small difference between GBW_{est} and GBW_{the}, which confirms that the non-dominant poles are sufficiently far from GBW as stated in Table 4.1.

We can now plot the gain response using the variables given in Table 4.1. It is shown in Figure 4.1.

4.2 Input-referred noise

We can now compute all the parameters needed for the calculation of the OTA thermal noise excess factor and its input-referred thermal noise PSD and resistance. They are given in Table 4.2.

© C. Enz under CC BY-NC-SA

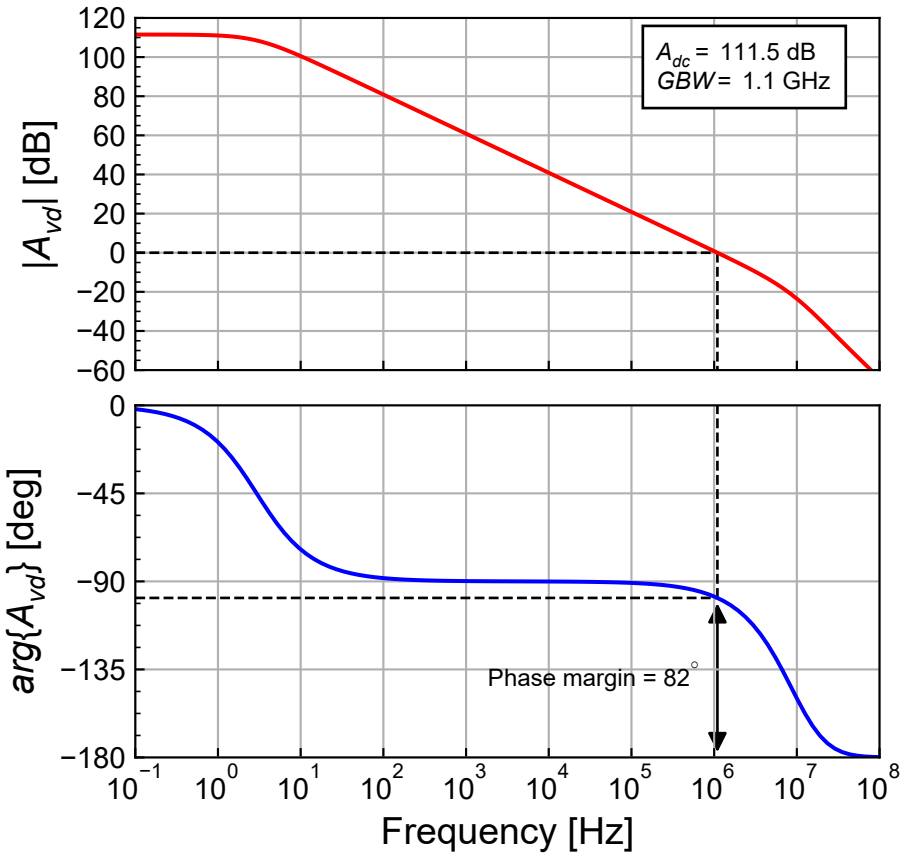


Figure 4.1: OTA theoretical transfer function.

Table 4.2: OTA thermal noise parameters.

Symbol	Theoretical Value	Unit
G_{m1}	6.964	$\mu A/V$
G_{m2}	1.384	$\mu A/V$
G_{m3}	1.783	$\mu A/V$
G_{m1}/G_{m2}	5.031	-
G_{m1}/G_{m3}	3.906	-
γ_{n1}	0.653	-
γ_{n2}	0.83	-
γ_{n3}	0.798	-
η_{th}	0.818	-
γ_{ota}	2.375	-
R_{nt}	341.056	$k\Omega$
S_{ninth}	5.6e-15	V^2/Hz
$\sqrt{S_{ninth}}$	75.152	$nV/\sqrt{Hz}$
$10 \cdot \log(S_{ninth})$	-142.481	$dBv/\sqrt{Hz}$

From the value of $\eta_{th} = 0.818$ in Table 4.2, we see that the current mirrors M_{2a}-M_{2b}-M_{2c}-M_{2d} and M_{3a}-M_{3b} contribute about equally to the OTA input-referred thermal noise PSD. The OTA thermal noise excess factor is therefore about twice that of the differential pair $2\,\gamma_{n1}$.

We can now compute all the parameters needed for the calculation of the input-referred flicker noise and the corner frequency. They are given in Table 4.3.

© C. Enz under CC BY-NC-SA

Table 4.3: OTA flicker noise parameters.

Symbol	Theoretical Value	Unit
$(G_{m1}/G_{m2})^2$	25.3	-
$(G_{m1}/G_{m3})^2$	15.3	-
ρ_p/ρ_n	8.3	-
$\frac{W_{eff1}\cdot L_{eff1}}{W_{eff2}\cdot L_{eff2}}$	1.2	-
$\frac{W_{eff1}\cdot L_{eff1}}{W_{eff3}\cdot L_{eff3}}$	0.9	-
η_{fl}	0.837	-
$\sqrt{S_{ninfl}(1\,Hz)}$	42	$\mu V/\sqrt{Hz}$
$10\cdot\log(S_{ninfl}(1\,Hz))$	-87.5	$dBv/\sqrt{Hz}$
f_k	311.9	kHz

Similarly to thermal noise, from the value of $\eta_{fl} = 0.837$ in Table 4.3, we see that the current mirrors M_{2a}-M_{2b}-M_{2c}-M_{2d} and M_{3a}-M_{3b} contribute about equally to the OTA input-referred flicker noise PSD, despite the large $\rho_p/\rho_n = 8.333$ ratio and the smaller area of the current mirrors compared to the differential pair. This is achieved because of the high gains $(G_{m1}/G_{m2})^2 = 25.3$ and $(G_{m1}/G_{m3})^2 = 15.3$.

We can plot the input-referred noise PSD which is shown in Figure 4.2. The corner frequency is estimated at $f_k = 311.9\ kHz$.

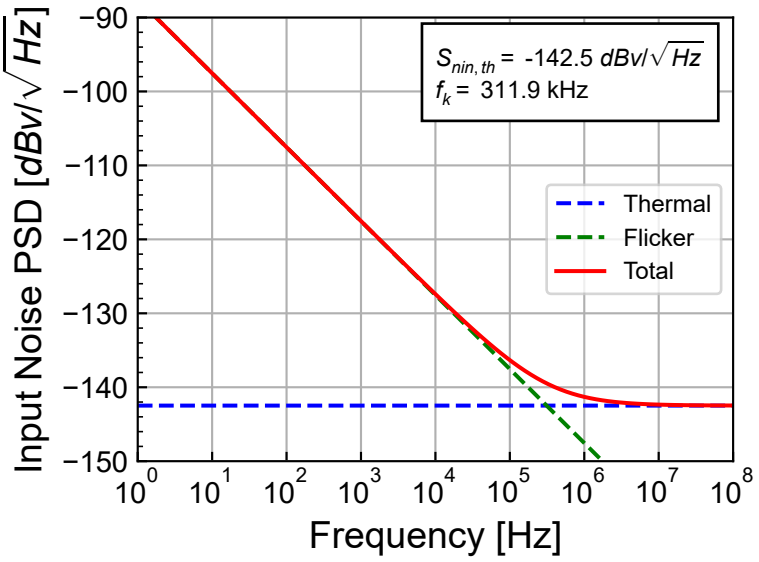


Figure 4.2: OTA theoretical input-referred noise PSD.

4.3 Input-referred offset voltage

The variance of the input-referred offset for $A_1 = A_2 = A_3 = 1$ is given by (2.45) which can be rewritten as

$$\sigma^2_{V_{os}} = \sigma^2_{V_T} + \sigma^2_{\beta} \tag{4.1}$$

where

$$\sigma^2_{V_T} = \sigma^2_{V_{T1}} \cdot (1 + \xi_{V_T}) \tag{4.2}$$

© C. Enz under CC BY-NC-SA

and

$$\sigma_\beta^2 = \left(\frac{I_b}{G_{m1}}\right)^2 \cdot \sigma_{\beta_1}^2 \cdot (1 + \xi_\beta). \tag{4.3}$$

ξ_{V_T} represents the V_T-mismatch contributions to the input-referred offset of the current mirrors relative to that of the differential pair. It is given by (2.54) which is repeated here

$$\xi_{V_T} = 2\left(\frac{G_{m2}}{A_1\, G_{m1}}\right)^2 \cdot \left(\frac{A_{VTp}}{A_{VTn}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \left(\frac{G_{m3}}{G_{m1}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff3}\, L_{eff3}} \tag{4.4}$$

and ξ_β represents the β-mismatch contributions to the input-referred offset of the current mirrors relative to that of the differential pair. It is given by (2.55) which is repeated below

$$\xi_\beta = 2\left(\frac{A_{\beta p}}{A_{\beta n}}\right)^2 \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff2}\, L_{eff2}} + \frac{1}{A_1^2} \cdot \frac{W_{eff1}\, L_{eff1}}{W_{eff3}\, L_{eff3}}. \tag{4.5}$$

with

$$\sigma_{V_{T1}}^2 = \frac{A_{VTn}^2}{W_{eff1}\, L_{eff1}}, \tag{4.6}$$

$$\sigma_{V_{T2}}^2 = \frac{A_{VTp}^2}{W_{eff2}\, L_{eff2}}, \tag{4.7}$$

$$\sigma_{V_{T3}}^2 = \frac{A_{VTn}^2}{W_{eff3}\, L_{eff3}}, \tag{4.8}$$

and

$$\sigma_{\beta 1}^2 = \frac{A_{\beta n}^2}{W_{eff1}\, L_{eff1}}, \tag{4.9}$$

$$\sigma_{\beta 2}^2 = \frac{A_{\beta p}^2}{W_{eff2}\, L_{eff2}}, \tag{4.10}$$

$$\sigma_{\beta 3}^2 = \frac{A_{\beta n}^2}{W_{eff3}\, L_{eff3}}. \tag{4.11}$$

The parameter for calculating the standard deviation of the input-referred offset voltage are given in Table 4.4.

Table 4.4: OTA input-referred offset parameters.

Symbol	Theoretical Value	Unit
σ_{VT1}	3.397	mV
σ_{VT2}	4.188	mV
σ_{VT3}	3.612	mV
$\sigma_{\beta 1}$	0.7	%
$\sigma_{\beta 2}$	0.8	%
$\sigma_{\beta 3}$	0.7	%
ξ_{VT}	0.194	-
ξ_β	4.17	-
σ_{VT}^2	13.782	mV^2
σ_{VT}	3.712	mV
σ_β^2	0.3	mV^2
σ_β	0.6	mV
σ_{Vos}	3.754	mV

From Table 4.4, we see that the β-mismatch is negligible and that the input-referred offset voltage is dominated by the contribution of the V_T-mismatch from the differential pair.

© C. Enz under CC BY-NC-SA

4.4 Current and power consumption

The total current consumption, ignoring the current drawn by M_{5a}, is $I_{tot} = (2+A_1+A_3)\cdot I_b$. Assuming the input differential pait M_{1a}-M_{1b} is biased in weak inversion, the bias current is directly related to the gain-bandwidth product GBW according to

$$I_b \cong nU_T \cdot \frac{C_L}{A_3} \cdot GBW. \tag{4.12}$$

The minimum total current consumption can then be estimated as

$$I_{tot,min} \cong \frac{2 + A_1 + A_3}{A_3} \cdot nU_T \cdot C_L \cdot GBW. \tag{4.13}$$

The bias current and the total current are proportional to the gain-bandwidth product GBW and the load capacitance C_L. In the case $A_1 = A_2 = A_3 = 1$, we have

$I_{tot,min} \cong 4 \cdot nU_T \cdot C_L \cdot GBW = 827\ nA.$

The actual current consumption accounting for some margin taken on the GBW is given by $I_{tot} = 4I_b = 1.00\ \mu A$ which is 1.210 larger than the absolute minimum current consumption. This larger current consumption is mostly due to the parasitic capacitances at the output which add to the exsiting load capacitance and require a larger transconductance and hence a larger current to achive the same GBW.

The above design will now be checked against simulations.

© C. Enz under CC BY-NC-SA

5 Simulation results from ngspice

The theoretical results can be validated with the results obtained from simulations performed with ngspice. In order to run the simulations you need to have ngspice correctly installed. Please refer to the installation instructions.

> **Note**
>
> The simulations are performed with ngspice [5] using the EKV 2.6 compact model [6] [13] [2]. For ngspice, we use the original Verilog-A implementation of EKV 2.6 [8] modified by C. Enz to get the operating point informations and available on the Gitub va-models site provided by D. Warning at [7] [14]. The Verilog-A code was then compiled with OpenVAF [15] to generate the OSDI for running it with ngspice. The parameters correspond to a generic 180 nm bulk CMOS process [9].

5.1 Operating point

We first write the parameter file for this specific design for running the ngspice simulations. Before running the AC and NOISE simulations, we first need to check the quiescent voltages and currents and the operating points of all transistors by running a .OP simulation.

Table 5.1: OTA node voltages with the OTA in open-loop without offset correction.

Node	Voltage
vdd	1.8
vb1	0.7
vb2	0.9
inp	0.9
inn	0.9
out	0.346562
ic	0.9
id	0
1	0.486491
2	0.971233
3	0.720522
4	0.971233
5	0.335405
6	1.26573
8	0.721661

From the node voltages shown in Table 5.1, we see that the output voltage $V_{outq} = 347\ mV$ is too low and is pushing M_7 and M_{3b} out of saturation. This is due to the current mismatch in M_{2c} and M_{2d} and between M_{3a} and M_{3b} due to different drain voltages and finite output conductances leading to a different current flowing in M_{2d} and M_4 compared to that flowing in M_{3b} and M_7. Even if the current

difference is very small, it results in a large voltage debiasing at the output because of the high output resistance provided by the cascode stages.

Therefore simulating the open-loop gain for high gain amplifiers is not easy to perform without closing the loop. There are basically two approaches to simulate the open-loop gain for high-gain amplifiers:

1) Imposing a DC offset voltage to the amplifier in open-loop configuration that brings the output voltage back to the high gain region (for example equal to the input common-mode voltage) or
2) Simulating the closed-loop gain (for example in voltage follower mode with a feedback gain of 1 and extracting the open-loop gain from the closed loop gain according to

$$A_{open\text{-}loop}(\omega) = \frac{A_{closed\text{-}loop}(\omega)}{1 - A_{closed\text{-}loop}(\omega)}, \tag{5.1}$$

where $A_{closed\text{-}loop}(\omega)$ is the simulated closed-loop transfer function and $A_{open\text{-}loop}(\omega)$, the computed open-loop transfer function. The above relation assumes that the open-loop DC gain is large enough for the input-referred offset voltage to be ignored.

The input-referred offset voltage can be extracted from the closed-loop voltage follower circuit as

$$V_{os} = V_{in} - \left(1 + \frac{1}{A}\right) \cdot V_{out} \cong V_{in} - V_{out} \quad \text{for } A \gg 1, \tag{5.2}$$

where $A \triangleq A_{open\text{-}loop}(0)$ is the open-loop DC gain which can be assumed to be much larger than 1. This means that, provided the DC open-loop gain is sufficiently large, the offset voltage can be measured at the amplifier differential input after imposing the proper input common-mode voltage V_{ic}.

We can now simulate the OTA in closed-loop as a voltage follower.

Table 5.2: OTA node voltages with the OTA in voltage follower configuration.

Node	Voltage
vdd	1.8
vb1	0.7
vb2	0.9
inp	0.9
out	0.899973
1	0.486477
2	0.971307
3	0.720468
4	0.971159
5	0.37758
6	1.26874
8	0.721661

From the node voltages of the OTA in voltage follower configuration shown in Table 5.2, we see that now the quiescent ouput voltage $V_{outq} = 900\ mV$ is very close to the input voltage that has been set to $V_{ic} = 900\ mV$. We can then extract the corresponding offset voltage $V_{os} \cong V_{in} - V_{out} = 27.000\ \mu V$.

We can now apply this offset voltage to the open-loop operating point point simulation.

Table 5.3: OTA node voltages with the OTA in open-loop including offset correction.

Node	Voltage
vdd	1.8

© C. Enz under CC BY-NC-SA

Table 5.3: OTA node voltages with the OTA in open-loop including offset correction.

Node	Voltage
vb1	0.7
vb2	0.9
inp	0.900014
inn	0.899987
out	1.05158
ic	0.9
id	2.7e-05
1	0.486491
2	0.971308
3	0.720467
4	0.971157
5	0.378381
6	1.26967
8	0.721661

From the node voltages of the OTA in open-loop configuration shown in Table 5.3 after adding the offset voltage at the input, we see that now the quiescent ouput voltage $V_{outq} = 1052\ mV$ of the open-loop circuit is sufficiently close to the common-mode input voltage $V_{ic} = 900\ mV$.

The operating point information for all transistors coming from the EKV2.6 compact model are extracted from the ngspice .op.dat file. The data is split into the large-signal operating informations presented in Table 5.4, the small-signal operating point informations shown in Table 5.5 and the noise operating point informations in Table 5.6.

We can compare the results of the .op file (for example the inversion coefficient IC) to the results of the design given in Table 3.7. We observe that the values are close.

Similarly we can compare the small-signal parameters (for example the gate transconductance G_m) resulting from the .op file to the results of the design presented in Table 3.8. Again, we see that they are reasonably close.

Table 5.4: Operating point information extracted from ngspice .op file for each transistor.

Transistor	$I_D\ [nA]$	$I_{spec}\ [nA]$	IC [-]	n [-]	$V_{DSsat}\ [mV]$
M1a	249.5	2569.1	0.097	1.27	120
M1b	249.7	2569.1	0.097	1.27	120
M2a	249.5	9.0	27.678	1.31	376
M2b	249.7	9.0	27.700	1.31	376
M2c	249.7	9.0	27.682	1.31	376
M2d	249.4	9.0	27.695	1.31	376
M3a	249.7	16.1	15.506	1.27	307
M3b	249.4	16.1	15.501	1.27	307
M4	249.4	2053.7	0.122	1.31	122
M5a	500.0	32.7	15.335	1.27	306
M5b	499.1	32.7	15.329	1.27	306
M7	249.4	2396.5	0.104	1.27	120

© C. Enz under CC BY-NC-SA

Table 5.5: Small-signal operating point information extracted from ngspice .op file for each transistor.

Transistor	G_m $[\mu A/V]$	G_{ms} $[\mu A/V]$	G_{ds} $[nA/V]$
M1a	6.839	8.828	17.666
M1b	6.844	8.834	17.681
M2a	1.212	1.643	0.973
M2b	1.212	1.644	0.974
M2c	1.213	1.645	0.873
M2d	1.211	1.642	1.152
M3a	1.643	2.117	0.765
M3b	1.640	2.114	2.372
M4	6.900	8.686	56.781
M5a	3.300	4.251	3.100
M5b	3.295	4.245	4.338
M7	7.076	8.728	43.947

Table 5.6: Noise operating point information extracted from ngspice .op file for each transistor.

Transistor	R_n $[k\Omega]$	$\sqrt{S_{ID,th}}$ $[nA/\sqrt{Hz}]$	γ_n [-]	$\sqrt{S_{ID,fl}}$ at 1Hz $[\mu A/\sqrt{Hz}]$
M1a	98.4	40.4	0.673	21.9
M1b	98.3	40.4	0.673	21.9
M2a	732.2	110.2	0.887	68.8
M2b	732.0	110.2	0.887	68.8
M2c	731.5	110.1	0.887	68.8
M2d	733.0	110.2	0.887	68.8
M3a	508.3	91.8	0.835	20.5
M3b	509.6	91.9	0.836	20.5
M4	95.1	39.7	0.656	132.0
M5a	253.8	64.9	0.838	29.3
M5b	254.3	64.9	0.838	29.3
M7	91.2	38.9	0.646	58.4

We can also check the bias voltages and operating region of each transistor which are given in Table 5.7.

Table 5.7: Bias voltages and operating regions extracted from ngspice for each transistor.

Trans.	Type	Funct.	V_G $[V]$	V_S $[V]$	V_D $[V]$	V_{DS} $[mV]$	V_{DSsat} $[mV]$	Reg.	Sat.
M1a	n	DP	0.414	0.000	0.485	485	120	WI	sat
M1b	n	DP	0.413	0.000	0.485	485	120	WI	sat
M2a	p	CM	0.829	0.000	0.829	829	376	SI	sat
M2b	p	CM	0.829	0.000	0.829	829	376	SI	sat
M2c	p	CM	0.829	0.000	1.080	1080	376	SI	sat
M2d	p	CM	0.530	0.000	0.530	530	376	SI	sat
M3a	n	CM	0.720	0.000	0.720	720	307	SI	sat
M3b	n	CM	0.720	0.000	0.378	378	307	SI	sat
M4	p	CA	1.100	0.530	0.748	218	122	MI	sat
M5a	n	CM	0.722	0.000	0.722	722	306	SI	sat
M5b	n	CM	0.722	0.000	0.486	486	306	SI	sat
M7	p	CA	0.900	0.378	1.052	673	120	MI	sat

© C. Enz under CC BY-NC-SA

From Table 5.4, we see that all transistors have a sufficiently large V_{DS} voltage and are therefore biased in saturation. Additionnally we see that all the saturation voltages in the output current branch M_{2d}, M_4, M_7 and M_{3b} are summing up to 924 mV leaving an output voltage swing about equal to $V_{out,swing} \cong 876\ mV$. This output voltage swing could be increased by about 250 mV by biasing M_{2d} and M_{3b} in weak inversion. This would be at the cost of larger transistors and a less current matching in the current mirrors.

From Table 5.7, we see that all transistors are biased in saturation. The operating points look fine. We can now proceed with the simulation of the open-loop large-signal transfer characteristic.

5.2 Large-signal differential transfer characteristic

We now simulate the DC differential transfer characteristic. We can then check the systematic offset voltage that was extracted above. The simulation of the large-signal input-output characteristic is presented in Figure 5.1.

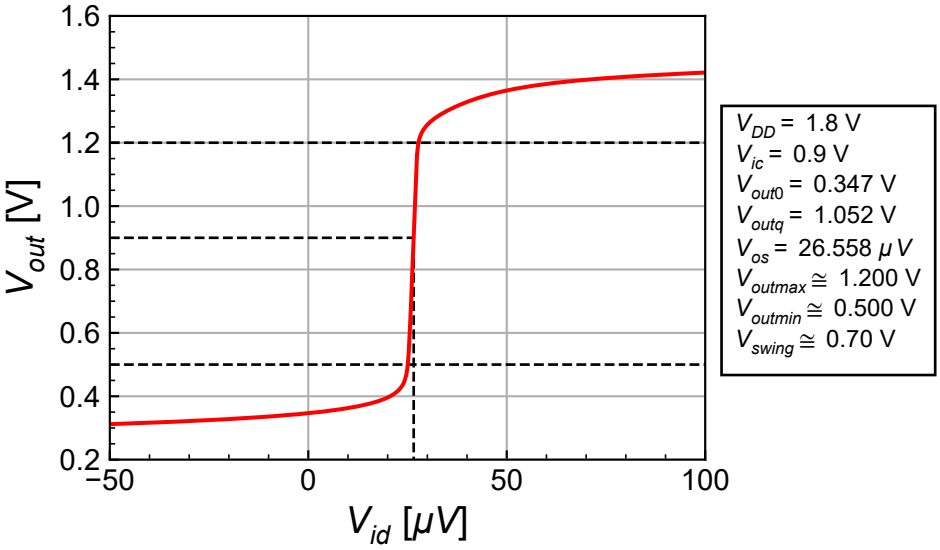


Figure 5.1: Simulated large-signal input-output characteristic.

From Figure 5.1, we see that the output swing is about $V_{out,swing} \cong 700\ mV$, which is slightly less than the above estimation. We can now zoom into the high gain region in order to extract the offset voltage that is needed to bring the output voltage back to $V_{outq} = V_{ic} = 0.900\ V$. The simulation results are presented in Figure 5.2.

From Figure 5.2 we also see that the linear region, where the gain is constant, is actually smaller resulting in a smaller output voltage swing $V_{out,swing} \cong 500\ mV$.

We can now save a more accurate value of the offset voltage $V_{os} = 26.556\ \mu V$ that is required to bring the output voltage to $V_{ic} = 900\ mV$ and that will be used for the following .AC and .NOISE simulations.

© C. Enz under CC BY-NC-SA

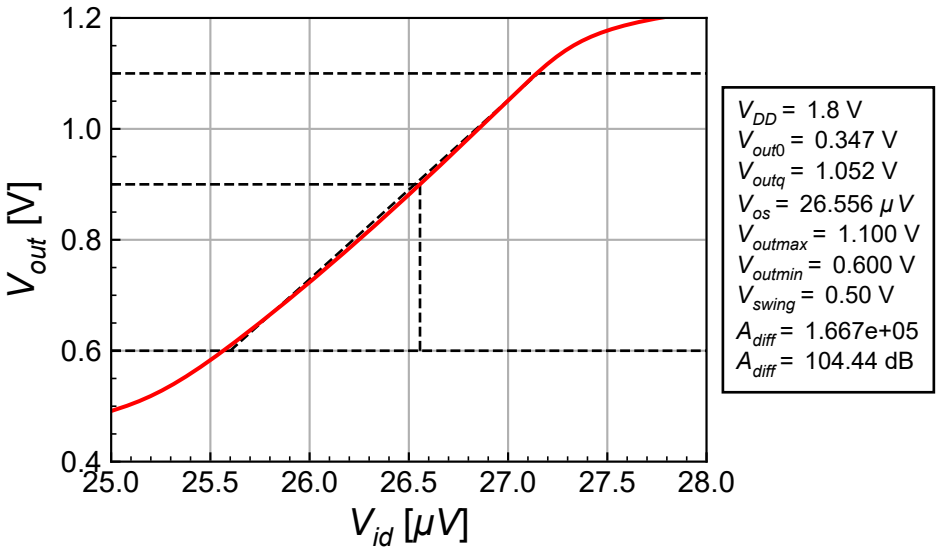


Figure 5.2: Zoom of the simulated large-signal input-output characteristic in the gain region.

5.3 Open-loop gain

5.3.1 Closed-loop circuit

As explained above we can extract the open-loop gain from the simulated closed-loop gain with the amplifier operating as a voltage follower. The open-loop gain is then given by

$$A_{open\text{-}loop}(\omega) = \frac{A_{closed\text{-}loop}(\omega)}{1 - A_{closed\text{-}loop}(\omega)}, \tag{5.3}$$

where $A_{closed\text{-}loop}(\omega)$ is the simulated closed-loop transfer function and $A_{open\text{-}loop}(\omega)$, the computed open-loop transfer function. The above relation assumes that the open-loop DC gain is large enough for the input-referred offset voltage to be ignored.

Note that we need to have a sufficient number of digits for the simulated closed-loop gain because the latter is very close to 1. The simulations results are shown in Figure 5.3.

From Figure 5.3, we see that the simulated GBW is right on target but slightly smaller than the estimated value. The simulated DC gain is much smaller than the theoretical estimation but slightly above spec. We can now simulate the open-loop gain directly in open-loop configuration and compare the results to those achieved from the closed-loop simulation.

5.3.2 Open-loop circuit

After having checked the operating point information, extracted the offset voltage and making sure that the OTA output is not saturated, we can now proceed with the open-loop gain simulation. The simulation results are presented in Figure 5.4.

The results of the open-loop simulation in Figure 5.4 are very close to the result extracted from the closed-loop simulation except at high frequency (above the GBW frequency). The GBW is the same than the value extracted from the closed-loop simulation, smaller than the estimated one but still on target. The DC gain obtained from the open-loop simulation gain is slightly smaller than the value extracted from the closed-loop simulation and much smaller than the theoretical estimation, but still within spec. The low value of the DC gain is due to the high output conductance of nMOS transitors in this technology, while the difference between simulation and theoretical prediction is due to the poor output conductance model that is used in the design phase.

© C. Enz under CC BY-NC-SA

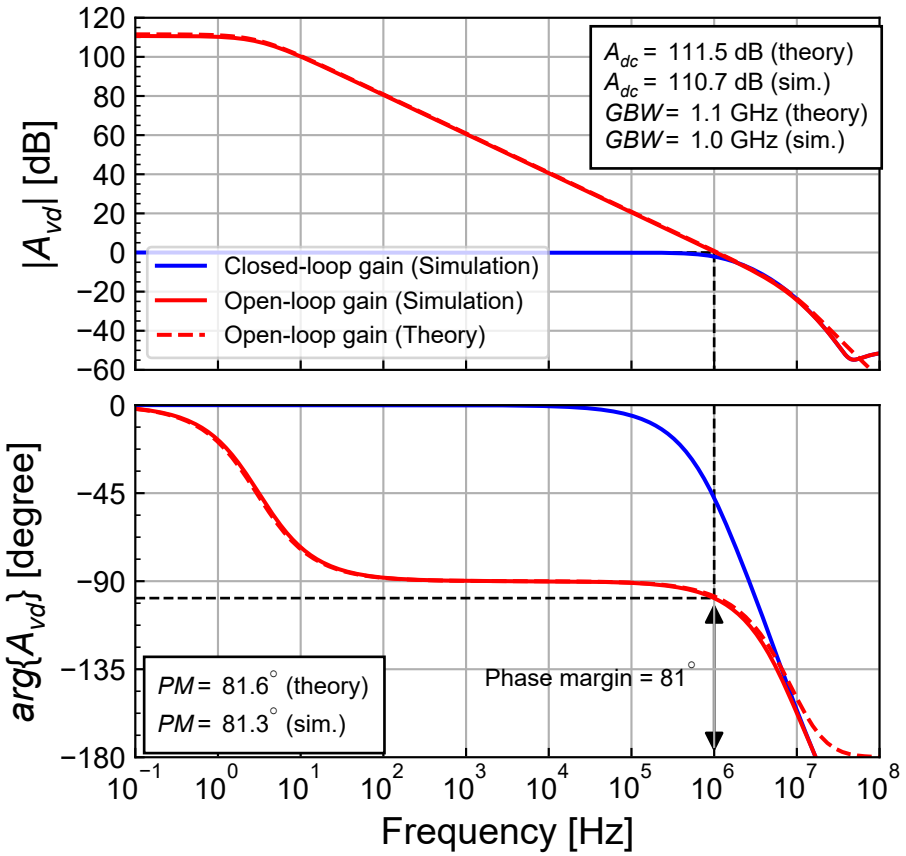


Figure 5.3: Open-loop gain response extracted from the closed-loop simulations and compared to theoretical estimation.

© C. Enz under CC BY-NC-SA

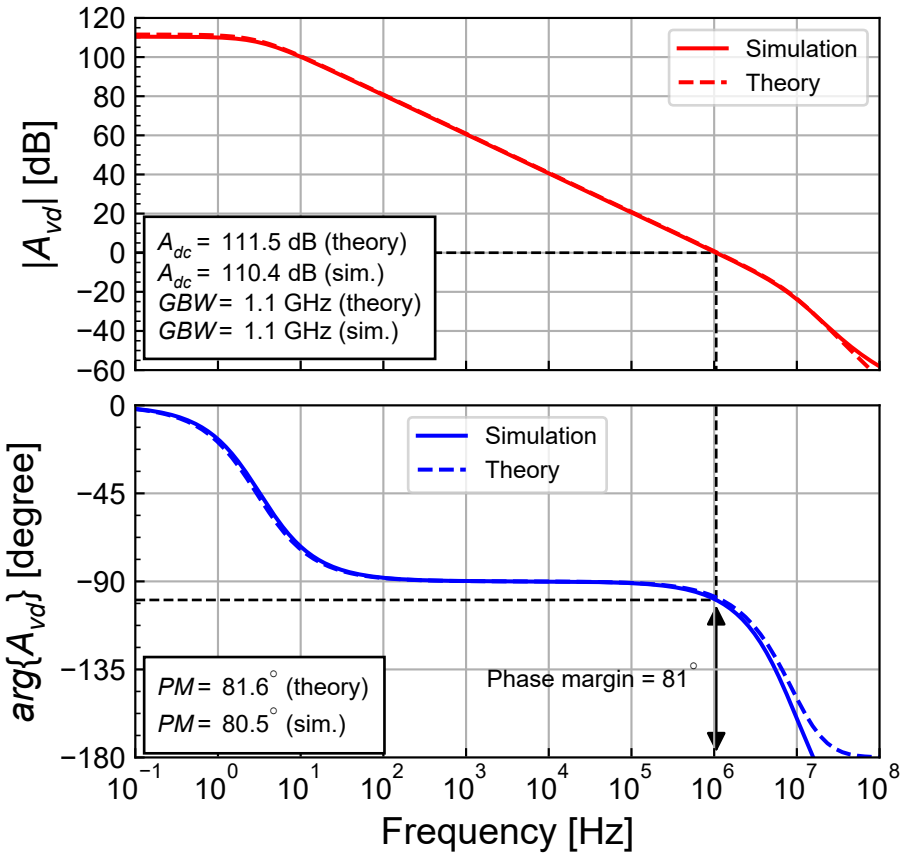


Figure 5.4: Simulated gain response compared to theoretical estimation.

5.4 Input-referred noise

The simulated input-referred noise PSD is presented in Figure 5.5 and compared to the theoretical estimation.

From Figure 5.5, we see a perfect match between simulation and theoretical estimation for both the flicker noise and the white noise except a^bove the GBW where the simulated noise PSD becomes larger than the theoretical estimation. This is simply due to the fact that the input-referred noise PSD has been obtain by dividing the output current noise by the square of the equivalent trnasconductance without accouting for the frequency dependence occuring above the GBW.

The contributions of M_{1a}-M_{1b}, M_{2a}-M_{2b}-M_{2c}-M_{2d}, M_{3a}-M_{3b} and the cascode transistors M_4 and M_7 to the input-referred white noise PSD are detailed in Figure 5.6 and compared to the theoretical white noise.

From Figure 5.6, we can observe a perfect match of the total input-referred white noise PSD between simulation and the theoretical estimation. The white noise is dominated by the differential pair M_{1a}-M_{1b} which is $1 + \eta_{th} = 1.782$ times (or 2.509 dB) lower than the total white noise. The contribution of M_{2a}-M_{2b}-M_{2c}-M_{2d} is about 2.107 times (or 3.237 dB) lower than the contribution of M_{1a}-M_{1b}. The contribution of M_{3a}-M_{3b} is about 3.255 times (or 5.125 dB) lower than the contribution of M_{1a}-M_{1b}. Finally, the contributions of the cascode transistors M_4 and M_7 are completely negligible. (80.517 dB lower for M_4 and 74.050 dB lower for M_7). This confirms that the noise of the cascode transistors M_4 and M_7 can be totally neglected.

The simulated value of $\eta_{th} = 0.782$ is almost equal to the theoretical estimation $\eta_{th} = 0.818$. Finally, the simulated OTA thermal noise excess factor $\gamma_{n,ota} = 2.403$ is slightly larger than the predicted value $\gamma_{n,ota} = 2.375$.

© C. Enz under CC BY-NC-SA

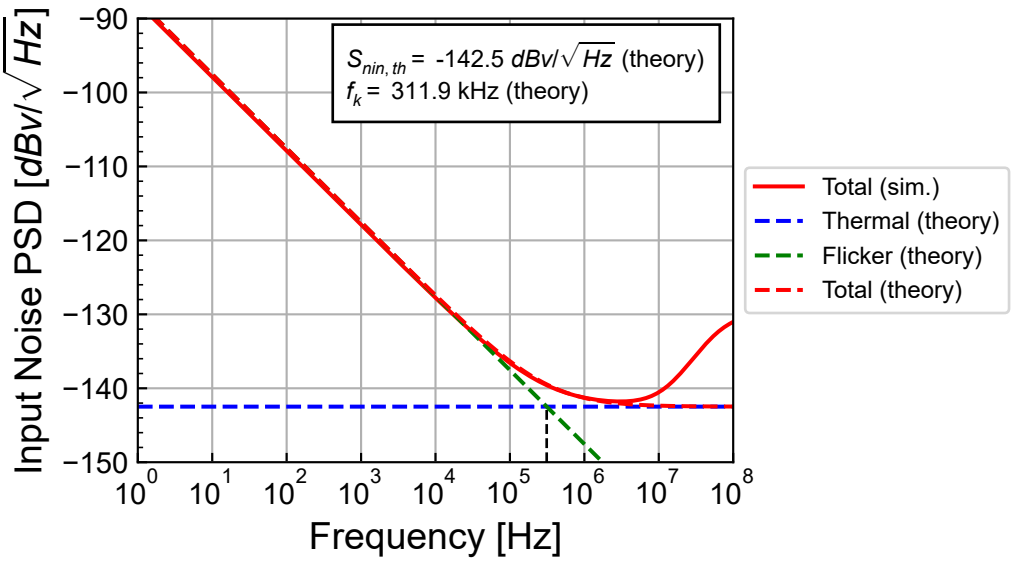


Figure 5.5: Simulated input-referred noise PSD compared to theoretical estimation.

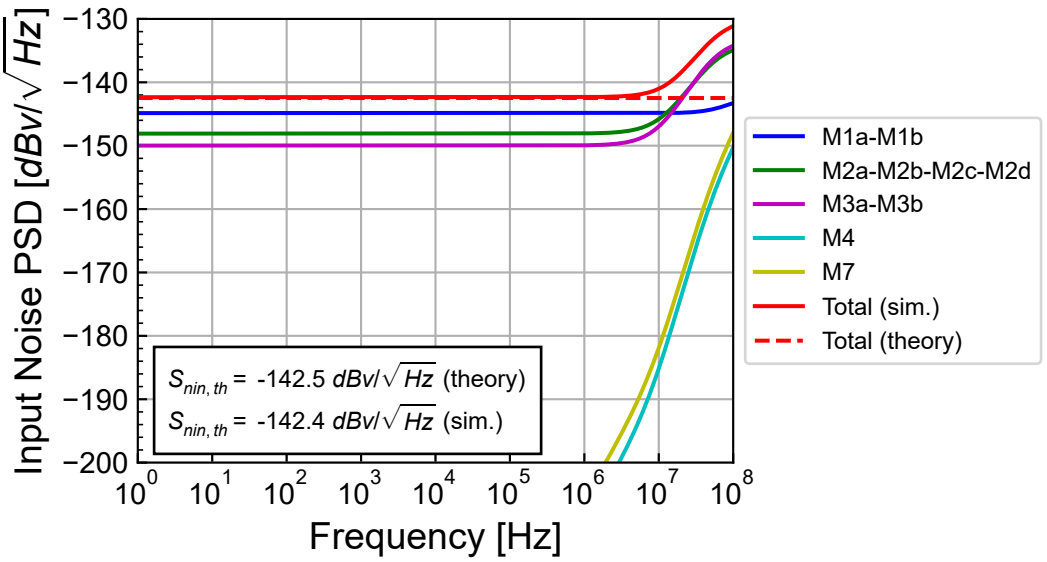


Figure 5.6: Breakdown of the contributions to the simulated input-referred white noise PSD.

© C. Enz under CC BY-NC-SA

Figure 5.7 presents the breakdown of the contributions of the various transistors to the input-referred flicker noise. As expected from the design, M_{2a}-M_{2b}-M_{2c}-M_{2d} contribute about the same than M_{1a}-M_{1b}.

The breakdown of the contributions of the various transistors to the total input-referred noise is presented in Figure 5.8. We can observe that the simulation is slightly larger than the theoretical estimation.

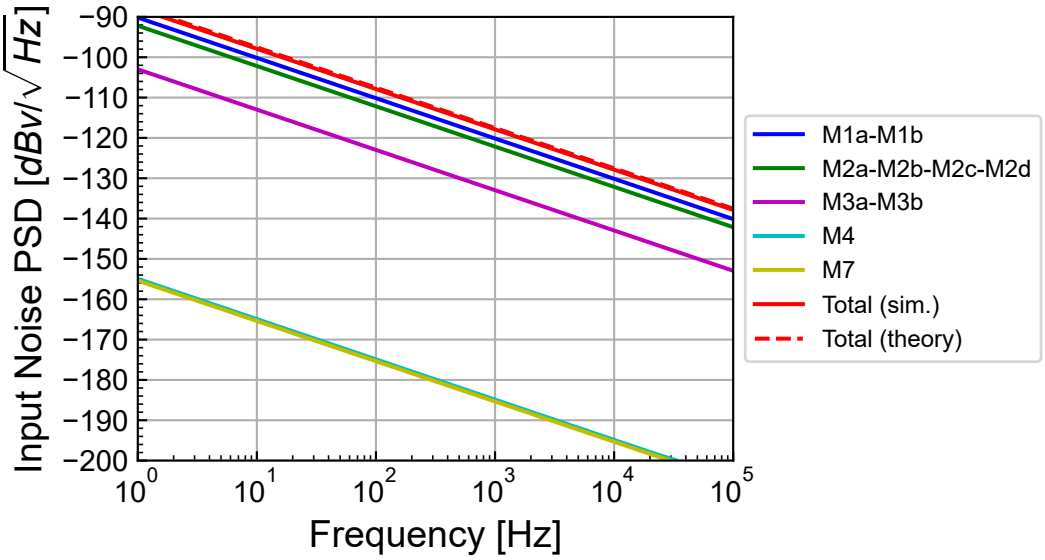


Figure 5.7: Breakdown of the contributions to the simulated input-referred flicker noise PSD.

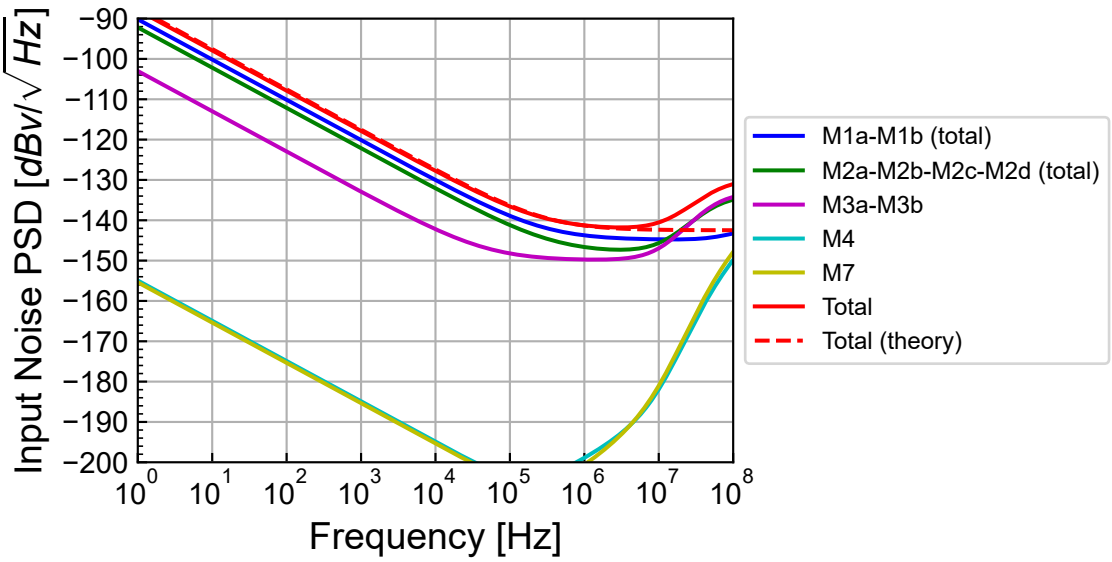


Figure 5.8: Breakdown of the contributions to the simulated input-referred noise PSD.

5.5 Input common-mode voltage range

We can check the input common-mode voltage range by connecting the OTA as a voltage follower and sweeping the positive input. The result is shown in Figure 5.9.

As shown in Figure 5.9, the output follows the input voltage up to 1.40 V. So the input common-mode voltage range is about 1.40 V.

© C. Enz under CC BY-NC-SA

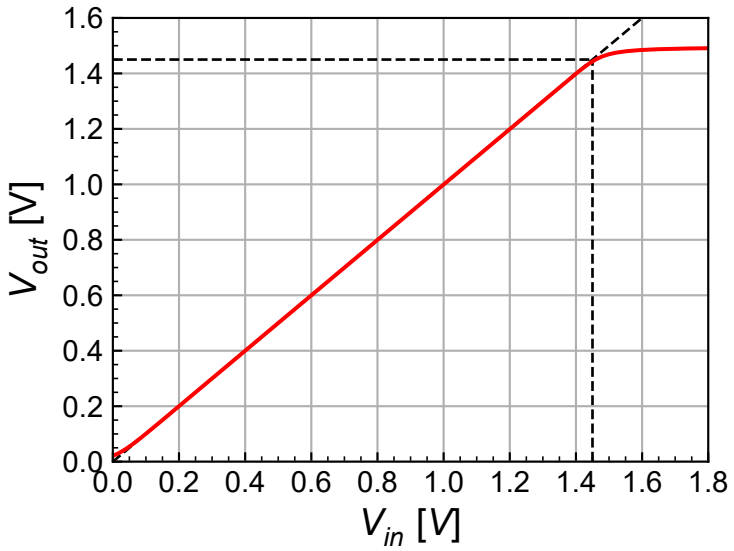


Figure 5.9: Simulated input common-mode voltage range.

5.6 Step-response

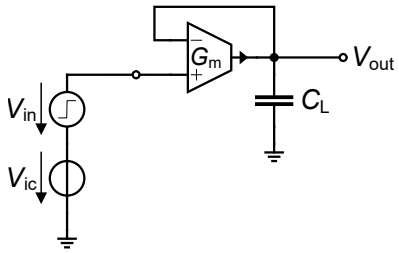


Figure 5.10: Schematic of the OTA connected as a voltage follower.

In this section we will check the step response of the OTA operating as a voltage follower as shown in Figure 5.10 with its output connected to the negative input and with the same load capacitance $C_L = 1\ pF$.

5.6.1 Small-step

We start by imposing a small step $\Delta V_{in} = 10\ mV$ on top of a common mode voltage $V_{ic} = 0.9\ V$. The simulation results are shown in Figure 5.11 where $\Delta V_{in}(t) \triangleq V_{in+}(t) - V_{ic}$ and $\Delta V_{out}(t) \triangleq V_{out}(t) - V_{outq}$ with V_{outq} the quiescent output voltage at the operating point before the step is applied. ΔV_{in} and ΔV_{out} are compared to the response of a single pole circuit having a cut-off frequency equal to the GBW. The difference between the simulation an the first-order model is due to the larger estimated value of the GBW and the additional poles introduced by the current mirrors.

5.6.2 Large step

We now impose a larger step $\Delta V_{in} = 300\ mV$ on top of a common mode voltage $V_{ic} = 400\ mV$. The simulation results are shown in Figure 5.12 where $\Delta V_{in}(t) \triangleq V_{in+}(t) - V_{ic}$ and $\Delta V_{out}(t) \triangleq V_{out}(t) - V_{outq}$ with $V_{outq} \cong V_{ic}$ the quiescent output voltage at the operating point before the step is applied. ΔV_{in} and ΔV_{out} are compared to the response of a single pole circuit having a cut-off frequency equal to the GBW. We now observe the effect of slew-rate which increases the settling time.

© C. Enz under CC BY-NC-SA

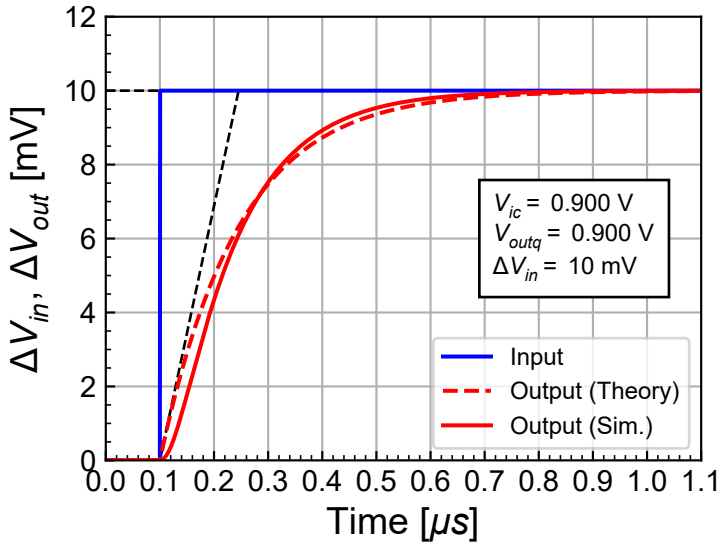


Figure 5.11: Step response of the OTA as a voltage follower for a small input step.

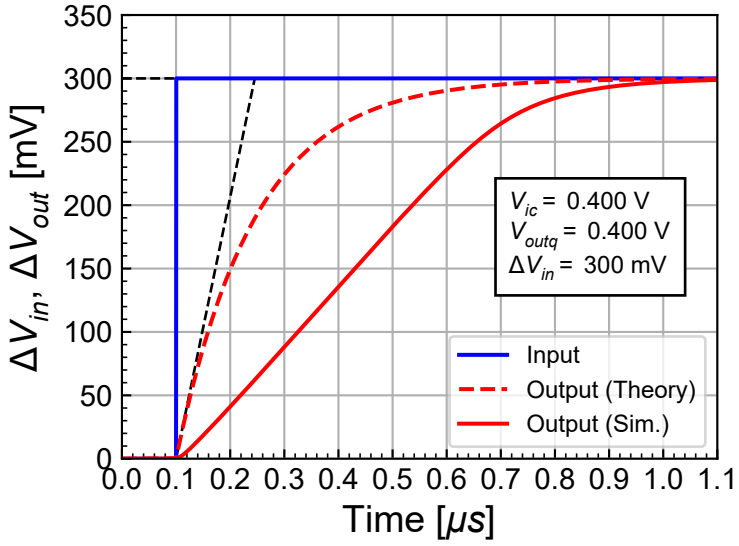


Figure 5.12: Step response of the OTA as a voltage follower for a large input step highlighting the slew-rate effect.

© C. Enz under CC BY-NC-SA

5.7 Current and power consumption

The total current consumption (without accounting for the current in M_{5a}) is given by $I_{tot} = 1.000$ μA, resulting in a total power consumption given by $P = 1.800$ μW. We can compare the current and power consumption of the Miller OTA to the telescopic OTA $I_{tot,telescopic} \cong 0.5$ μA. The current and power consumption of the symmetrical cascode OTA is 2.000 times larger than that of the telescopic OTA for similar specifications and performance.

© C. Enz under CC BY-NC-SA

6 Conclusion

This notebook presented the analysis, design and verification of the symmetrical cascode OTA [1] designed for a generic 180nm bulk CMOS process. The detailed analysis provided all the equations that were then used in the design phase to reach the target specifications. The design was then performed using the inversion coefficient approach with the sEKV transistor model [2] [3] [4]. The theoretical performance resulting from the design were then evaluated.

The design was then verified by simulation using ngspice [5] with the EKV 2.6 compact model [6] and the parameters of a generic 180 nm bulk CMOS process. Because of the high output resistance and the structural mismatch in the current mirrors due to the output conductances, the open-loop quiescent output voltage was too low, pushing some transistors out of saturation. The OTA was then simulated in closed-loop as a voltage follower in order to extract the offset voltage required to bring the quiescent output voltage in open-loop back to the high gain region. After carefully checking the operating point, the large-signal transfer characteristic was simulated. Then the small-signal transfer function of the OTA connected as a voltage follower was simulated. The open-loop transfer function was then simulated making sure the OTA was biased in the high gain region. The transfer function was then compared to the that extracted from the closed-loop simulation showing a reasonable correspondance below the unity gain frequency. The simulations have shown that the gain-bandwidth product GBW and the DC gain are both achieved.

The input-referred noise was then simulated showing results that are very close to the theoretical estimation for both the white noise and the flicker noise. The contributions of the various transistors to the input-referred white noise were then extracted from the noise simulation and compared to the theoretical estimation. It was shown that the total white noise is very close to the theoretical estimation. It was then shown that the flicker noise of the current mirrors is about equal to that of the differential pair. Finally, the simulations have confirmed that the noise contributions of the cascode transistors are completely negligible.

The input common-mode voltage range was then simulated with the OTA connected as a voltage follower.The input voltage is limited to $1.4\,V$.

Finally, the small-signal step response was simulated and succesfully compared to the response of a single-pole system. The step-response with a large input step highlighted the effect of slew-rate.

© C. Enz under CC BY-NC-SA

References

[1] F. Krummenacher, "High voltage gain CMOS OTA for micropower SC filters," *Electronics Letters*, vol. 17, no. 4, pp. 160–162, 1981.

[2] C. C. Enz and E. A. Vittoz, *Charge-Based MOS Transistor Modeling - The EKV Model for Low-Power and RF IC Design*, 1st ed. John Wiley, 2006.

[3] C. Enz, F. Chicco, and A. Pezzotta, "Nanoscale MOSFET Modeling: Part 1: The Simplified EKV Model for the Design of Low-Power Analog Circuits," *IEEE Solid-State Circuits Magazine*, vol. 9, no. 3, pp. 26–35, 2017.

[4] C. Enz, F. Chicco, and A. Pezzotta, "Nanoscale MOSFET Modeling: Part 2: Using the Inversion Coefficient as the Primary Design Parameter," *IEEE Solid-State Circuits Magazine*, vol. 9, no. 4, pp. 73–81, 2017.

[5] Holger Vogt, Giles Atkinson, Paolo Nenzi, "Ngspice User's Manual Version 43." https://ngspice.sourceforge.io/docs/ngspice-43-manual.pdf, 2024.

[6] M. Bucher, C. Lallement, C. Enz, F. Théodoloz, and F. Krummenacher, "The EPFL-EKV MOSFET Model Equations for Simulation." https://github.com/chrisenz/EKV/blob/main/EKV2.6/docs/ekv_v26_rev2.pdf, 1998.

[7] M. Bucher, C. Lallement, C. Enz, F. Théodoloz, and F. Krummenacher, "EKV 2.6 Verilog-A Code." https://github.com/chrisenz/EKV/tree/main/EKV2.6/va/code, 2024.

[8] W. Grabinski *et al.*, "FOSS EKV2.6 verilog-a compact MOSFET model," in *European solid-state device research conference (ESSDERC)*, 2019, pp. 190–193. doi: 10.1109/ESSDERC.2019.8901822.

[9] W. Grabinski *et al.*, "FOSS EKV 2.6 parameter extractor," in *2015 22nd international conference mixed design of integrated circuits & systems (MIXDES)*, 2015, pp. 181–186. doi: 10.1109/MIXDES.2015.7208507.

[10] Han, H.-C. and A. D'Amico and C. Enz, "SEKV-E: Parameter Extractor of Simplified EKV I-V model for Low-power Analog Circuits." https://gitlab.com/moscm/sekv-e, 2022.

[11] F. Krummenacher, E. Vittoz, and M. Degrauwe, "Class AB CMOS amplifier micropower SC filters," *Electronics Letters*, vol. 17, no. 13, pp. 433–435, 1981.

[12] M. G. Degrauwe, J. Rijmenants, E. A. Vittoz, and H. J. D. Man, "Adaptive biasing CMOS amplifiers," *IEEE Journal of Solid-State Circuits*, vol. 17, no. 3, pp. 522–528, 1982.

[13] C. C. Enz, F. Krummenacher, and E. A. Vittoz, "An Analytical MOS Transistor Model Valid in All Regions of Operation and Dedicated to Low-Voltage and Low-Current Applications," *Analog Integrated Circuits and Signal Processing Journal*, vol. 8. https://github.com/chrisenz/EKV/tree/main/EKV2.6/docs/EKV_original_paper_1995_prepub.pdf, pp. 83–114, 1995.

[14] Dietmar Warning, "Verilog-A Models for Circuit Simulation." https://github.com/dwarning/VA-Models, 2024.

[15] SemiMod GmbH, "Open VAF." https://openvaf.semimod.de/docs/getting-started/introduction/, 2025.

© C. Enz under CC BY-NC-SA